The Welfare Cost of Social Security Insolvency Risk

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This paper studies the welfare cost of perceived Social Security insolvency. If households expect future benefit cuts, they may save more, work longer, or claim earlier before any cut occurs. I quantify these distortions in a life-cycle model with endogenous saving, labor supply, and claiming, disciplined by 1995–2025 subjective expectations data and linked to OASI finances. The model measures private welfare losses and fiscal feedbacks, asking whether pessimism is self-validating, self-correcting, or mainly privately costly. The results show that credible communication about payable benefits can have value even without changing taxes or scheduled benefits.

Keywords: Social Security trust-fund insolvency, policy uncertainty, retirement

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1. Introduction

Social Security is the primary retirement income source for most of older Americans. However, its trust fund is projected for depletion in 2033, after which continuing payroll tax revenue would cover only about 77% of scheduled benefits (U.S. Social Security Administration 2025). The conventional belief that public entitlements are safe assets can miss a first-order component of trust fund accounting. If households anticipate insolvency, they act on it now, and the resulting distortions to how much they save, how long they work, and when they claim impose a welfare cost long before any benefit is actually cut, and whether a cut ever arrives.

This paper studies the economic cost of the gap between payable benefits and perceived benefits. The conventional accounting of Social Security insolvency locates the cost at the date of trust-fund exhaustion, when scheduled benefits can no longer be fully financed. I argue that this misses a distinct and potentially first-order cost. If households perceive future benefits as risky, they adjust well before any cut occurs. They may save more, work longer, or claim earlier to insure against the perceived loss of future benefits. These responses are privately costly because they distort consumption smoothing, labor supply, and the use of Social Security as longevity insurance. They are also fiscally relevant because claiming and labor supply affect the program's outlays and payroll-tax base. Therefore, expectations about the system's solvency are crucial determinants of financial status of the trust fund.

I quantify this mechanism in a partial-equilibrium life-cycle model with endogenous consumption, saving, labor supply, and Social Security claiming. The model differs from standard claiming models in its treatment of expectations. Households do not mechanically hold model-consistent beliefs about future policy. Instead, their perceived insolvency risk is disciplined by a 1995–2025 series of subjective Social Security expectations. These beliefs enter the household problem as a perceived risk that future scheduled benefits will be reduced, with the exposure to that risk allowed to depend on claiming age. Aggregating household decisions through the Old-Age & Survival Insurance (OASI) accounting identity generates payroll-tax revenue, benefit outlays, and the trust-fund path. This allows me to measure both the private welfare cost of perceived insolvency risk and the fiscal feedback from belief-driven behavior.

The mechanism is theoretically ambiguous. If households claim earlier because they believe current beneficiaries will be protected from future reforms, benefit outlays are pulled forward and the trust fund deteriorates more quickly. If they jointly decide to work longer or save more, payroll-tax revenue and private retirement resources may rise, partially offsetting the fiscal cost. The sign and magnitude of the feedback are therefore quantitative objects. The model allows me to ask the following: whether pessimism about Social Security has been self-validating with belief-driven claiming hastening the depletion households fear, or self-correcting with longer work and higher saving shoring up the program, or fiscally close to neutral leaving a private welfare loss as the main cost.

This paper's contribution is to treat Social Security insolvency not only as a future financing event but as a current expectation shock. Luttmer and Samwick (2018) has shown that perceived Social Security policy uncertainty carries welfare costs and that subjective expectations affect retirement planning. This paper adds the dynamic fiscal feedback. The perceived insolvency risk changes claiming, work, and saving decisions, and those decisions change the program finances that households are trying to forecast. The result is a new margin for policy. Credible communication about the payable-benefit floor, the incidence of future reforms, and the treatment of current beneficiaries can have economic value even without changing taxes or scheduled benefits.

Related literature. The first is the structural literature on endogenous retirement, which I extend by separating the claiming margin from the labor-supply margin. Life-cycle models of retirement behavior typically tie the two together or treat claiming as mechanical (Rust and Phelan 1997; French 2005; French and Jones 2011), while a parallel literature (Coile et al. 2002; Friedberg 2000; Gruber and Orszag 2003; Gelber, Jones, and Sacks 2020) studies the timing of benefit claims and the labor-supply response to the earnings test largely in isolation from each other. Pashchenko and Porapakkarm (2024) jointly model claiming and labor supply but within a stable, fully credible benefit schedule, and Bairoliya and McKiernan (2023) have households misperceive the existing early-claiming adjustment factor. My agents instead respond to the risk that the schedule itself is reformed—risk about the rules to come rather than misperception of current rules. Taking that misperception as a calibrated input, I ask what forward-looking insolvency risk adds to claiming, labor supply, welfare, and the trust fund. Because perceived insolvency risk can push claiming and work in opposite directions, modeling the two margins jointly but separately is essential.

The second is the literature on policy uncertainty and its measurement. Luttmer and Samwick (2018) use a cross-sectional survey of beliefs to show that perceived Social Security policy uncertainty carries a welfare cost. Manski (2004) provides the framework for using such expectations data directly. A macro literature instead builds aggregate indices of policy and economic uncertainty from text and forecasts (Baker, Bloom, and Davis 2016; Jurado, Ludvigson, and Ng 2015; Caldara and Iacoviello 2022; Bloom 2009). I differ from both: rather than a one-time survey or a text-based index, I construct a domain-specific, time-varying insolvency index from the 1995–2025 series of subjective Social Security expectations, and feed it into the model as the perceived benefit-cut risk that drives behavior and fiscal feedback.

The third is the literature that reframes public entitlements as risky rather than safe assets. A finance perspective treats accrued Social Security wealth as a risky contingent claim rather than a safe asset: Geanakoplos and Zeldes (2010) price benefits using derivative-pricing methods because the wage-indexed payout is correlated with the market and therefore carries a risk premium—a risk-adjusted valuation that Novy-Marx and Rauh (2011) apply to public pension promises more broadly—and Shoven and Slavov (2006) emphasize the political risk that future legislatures rewrite tax and benefit rules in response to demographic and fiscal pressure. Once benefits carry such risk, the household problem becomes one of precautionary self-insurance against an impaired future claim through saving and labor supply. In all of this work the riskiness of the entitlement is exogenous. My contribution is to make that riskiness endogenous. Aggregating belief-driven claiming, work, and saving through the OASI accounting identity, I let household responses to perceived insolvency feed back into payroll revenue, outlays, and the trust fund, which in turn determine actual solvency. Perceived and actual riskiness are thus jointly determined rather than taken as given. This is what lets me ask a question the earlier literature cannot: whether Social Security pessimism is self-validating, self-correcting, or primarily a private welfare loss.

Roadmap.

2. Measuring Social Security Insolvency Risk

The model requires a belief about whether scheduled Social Security benefits will be paid in full. The difficulty is that this belief has no market counterpart. Social Security wealth is non-tradable, there is no security whose price moves with the perceived odds of a future benefit cut, and households never observe a clean signal of program solvency—they observe a stream of official reports, projections, and political commentary that they must aggregate into a single sense of how safe the scheduled benefit is. Measuring an economic force that markets do not price is a recurring problem in macroeconomics, and the standard response has been to construct an index from the observable information set: Baker, Bloom, and Davis (2016) build economic policy uncertainty from newspaper text, Caldara and Iacoviello (2022) measure geopolitical risk the same way, and Jurado, Ludvigson, and Ng (2015) extract a common uncertainty factor from the forecast errors of many series. I follow this tradition and build a real-time *solvency-risk index* for Social Security, then anchor its scale to direct survey beliefs so that the abstract index becomes the subjective probability that enters the household problem.

The index uses what households, policymakers, and the media actually see, not realized benefit cuts—which never occurred in the sample and so cannot be used to date risk. Two economically distinct forces drive perceived insolvency risk, and the index keeps them separate. The first is *fiscal stress*, the level of the financing problem: a larger actuarial deficit, a larger unfunded obligation, a nearer depletion date, a lower payable benefit at depletion, and a weaker payroll base all mean the scheduled benefit is harder to finance and so more likely to be cut. Stress is a statement about the first moment—how bad the central projection is. The second force is *uncertainty*, the dispersion around that projection: wider Trustees scenario bands, larger stochastic dispersion, bigger release-to-release revisions, greater SSA–CBO disagreement, macro-data revision risk, and official text that emphasizes solvency uncertainty. This block captures the second moment, in the spirit of Jurado, Ludvigson, and Ng (2015) and Bloom (2009): even holding the expected funding gap fixed, a household that cannot pin down when or how the program will be adjusted faces more benefit risk, and is more likely to act on it. Separating the two blocks lets the data speak to which channel—the deteriorating level or the widening fan of outcomes—moves perceived risk over time.

I summarize each block with a single latent factor and combine them into the headline state G_t, placing most of the weight on stress and the remainder on uncertainty. The logic is that the level of the funding shortfall is the central object households are pricing, while disagreement, revisions, and official discussion shift how risky the benefit appears around that level. Figure 2 plots the two components. Two features of the construction matter for interpretation and are detailed in Appendix B.2, which gives the full data, sign conventions, standardization, and factor model. First, every input is dated by its release month and carried forward until new information arrives, so G_t is strictly real-time and never uses information households could not yet have seen. Second, the factor model lets a noisy or temporarily missing source drop out without contaminating the state, much as Jurado, Ludvigson, and Ng (2015) let a common factor filter idiosyncratic noise from many indicators.

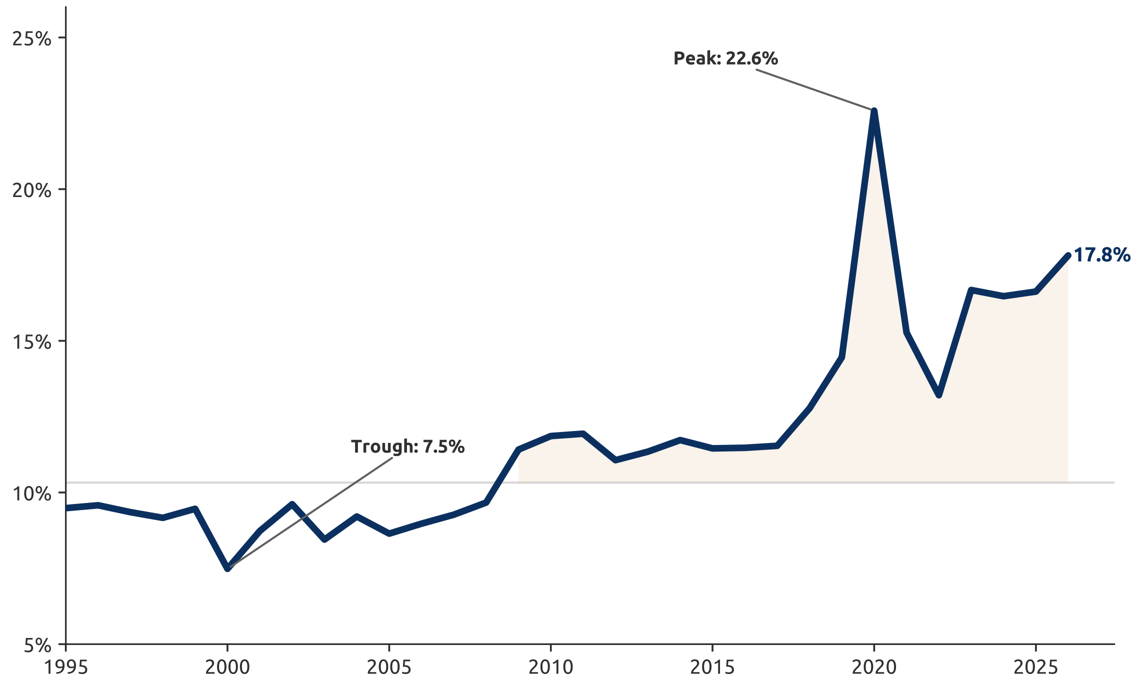


FIGURE 1. Subjective annual probability of Social Security insolvency

Note. The figure converts the annual solvency-risk state G_t into an annual subjective insolvency probability, $p_t = \Lambda(\alpha_0 + \alpha_1 G_t)$, where $\Lambda(\cdot)$ is the logistic function. The parameters α_0 and α_1 are estimated to match HRS Internet Survey responses on the probability that Social Security will become less generous over the next 10 years. The plotted object is the annual transition probability implied by those 10-year subjective beliefs.

A solvency-risk index, like an uncertainty index, is identified only up to scale: a standardized factor has no natural units, yet the household problem needs a probability. I therefore anchor G_t to elicited beliefs. The HRS Internet Survey asks respondents for the percent chance that Congress will change Social Security over the next 10 years so that the program becomes less generous, and I read these answers as subjective probabilities of a future policy-induced benefit reduction. Using survey responses to discipline a constructed index follows Manski (2004), who argues that subjective probabilities can be measured and used directly, and complements Luttmer and Samwick (2018), who document that perceived Social Security policy uncertainty carries welfare costs. The mapping fits a logistic hazard, $p_t = \Lambda(\alpha_0 + \alpha_1 G_t)$, comparing its implied 10-year probability to the HRS response: the survey moments pin down the level and slope of perceived risk, while the official index supplies the time-series variation that the cross-sectional survey cannot. The resulting series in Figure 1 is the annual subjective probability of insolvency that enters the household's perceived benefit process; the decomposition in Figure 2 is for interpretation only.

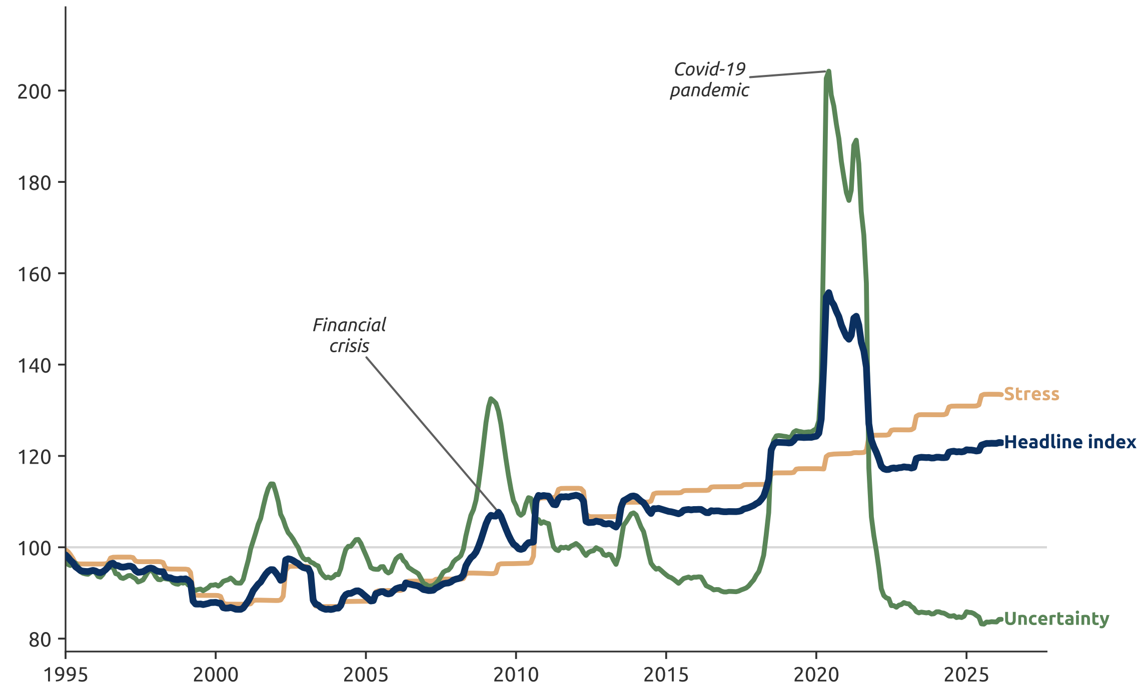


FIGURE 2. Stress and uncertainty components of the OASDI solvency-risk index

Note. The headline index is a weighted combination of a stress block and an uncertainty block. The stress block loads on official measures of the funding gap, depletion horizon, payable-benefit shortfall, trust-fund weakness, and payroll-base stress. The uncertainty block loads on scenario dispersion, revision volatility, SSA-CBO disagreement, macro revision risk, and the validated text-based uncertainty signal. Each series is normalized relative to the 1995–2019 baseline.

3. Simple Model

Before turning to the full life-cycle model, I use a two-period model to isolate the mechanism. The household is near retirement and faces a simple tradeoff. It can work and save today to build liquid private wealth, and it can either claim Social Security now or wait for a larger future benefit. If Social Security were safe, the future benefit would be just another source of retirement income. If households perceive insolvency risk, however, the future benefit becomes a risky claim that cannot be sold, borrowed against, or hedged. The only way to reduce this exposure is to work more to self-insure, or claim earlier to convert part of the risky future benefit into a current payment. The simple model shows why both option values rise when the perceived probability of insolvency increases.

3.1. Environment

The household starts with liquid wealth x, earns wage w, chooses labor $n \geq 0$, and saves $a \geq 0$ in a safe asset with gross return R. Preferences are

$$u(c_1) - v(n) + \beta\, \mathbb{E} u(c_2), \qquad u' > 0, \quad u'' < 0, \quad v' > 0, \quad v'' > 0.$$

The household chooses between two claiming options. If it claims early, it receives benefit b in period 1. If it delays, it receives the larger scheduled benefit $B > b$ in period 2. Future benefits are risky: with perceived probability $p \in [0, 1]$, insolvency reduces the delayed benefit by fraction $\delta \in (0, 1)$. Thus the delayed benefit is B with probability $1 - p$ and $(1 - \delta)B$ with probability p. I assume the maximization problems below have solutions over feasible choices with positive consumption.

The value of claiming early is

$$V_E = \max_{n \geq 0,\ a \geq 0} \left\{ u(x + wn + b - a) - v(n) + \beta u(Ra) \right\}.$$

The value of delaying is

$$\begin{aligned} V_D(p) = \max_{n \geq 0,\ a \geq 0} \Big\{ & u(x + wn - a) - v(n) \\ & + \beta \left[(1 - p) u(Ra + B) + p\, u(Ra + (1 - \delta)B) \right] \Big\}. \end{aligned}$$

This is the simple-model analog of the full model's perceived benefit process: higher p lowers the expected value of future Social Security wealth and raises the value of resources in the bad retirement state.

3.2. Option Values

First consider the value of working more when the household delays. For a given asset position, an additional dollar earned and saved raises second-period resources in both benefit states. The marginal value of this extra work is

$$\Omega^W(p) = w\beta R \left[(1 - p) u'(Ra + B) + p\, u'(Ra + (1 - \delta)B) \right].$$

PROPOSITION 1 (Work option value). *For any fixed feasible asset position a, the option value of working more, $\Omega^W(p)$, is strictly increasing in the perceived insolvency probability p.*

Proof. For fixed a, $\Omega^W(p)$ is differentiable in p. Its derivative is

$$\frac{d\Omega^W(p)}{dp} = w\beta R\left[u'(Ra + (1-\delta)B) - u'(Ra + B)\right].$$

Since $\delta B > 0$, the bad-state consumption level $Ra + (1-\delta)B$ is strictly below the good-state consumption level $Ra + B$. Strict concavity of u implies that u' is strictly decreasing, so $u'(Ra + (1-\delta)B) > u'(Ra + B)$. With $w > 0$, $\beta > 0$, and $R > 0$, the derivative is strictly positive. Hence $\Omega^W(p)$ is strictly increasing in p. □

Now consider the value of claiming early rather than delaying,

$$\Omega^C(p) = V_E - V_D(p).$$

PROPOSITION 2 (Early-claiming option value). *The option value of claiming early, $\Omega^C(p)$, is increasing in the perceived insolvency probability p.*

Proof. Let $F_D(n, a; p)$ denote the objective inside $V_D(p)$. Take any $p_2 > p_1$. For any feasible pair (n, a),

$$F_D(n, a; p_2) - F_D(n, a; p_1) = \beta(p_2 - p_1)\left[u(Ra + (1-\delta)B) - u(Ra + B)\right] < 0,$$

because u is strictly increasing and $Ra + (1-\delta)B < Ra + B$. Let (n_2, a_2) be an optimizer at p_2. Then

$$V_D(p_2) = F_D(n_2, a_2; p_2) < F_D(n_2, a_2; p_1) \leq V_D(p_1).$$

Therefore $V_D(p)$ is strictly decreasing in p. Since V_E does not depend on p, $\Omega^C(p) = V_E - V_D(p)$ is strictly increasing in p.

At points where V_D is differentiable, the same result can be written with the envelope theorem as

$$\frac{d\Omega^C(p)}{dp} = \beta\left[u(Ra_D + B) - u(Ra_D + (1-\delta)B)\right] > 0,$$

where a_D is the saving choice under delay. □

Higher p lowers the value of delaying because it puts more probability weight on the cut-benefit state. Claiming early is valuable because it converts part of a risky, non-tradable future claim into a current payment. Thus the same perceived insolvency risk raises both option values: the value of working more to self-insure and the value of claiming early to reduce exposure to future benefit risk.

This formulation makes the channel stark by treating the current early benefit as protected from the future insolvency draw. Allowing early claimants to remain partially exposed gives the same result as long as delaying carries the larger bad-state utility loss, which is natural when the delayed benefit is both larger and more exposed to future reform risk.

3.3. What the Simple Model Captures and What It Misses

The value of this stripped-down model is that it isolates one mechanism cleanly. A single perceived risk—the probability p that future scheduled benefits are cut—raises two distinct margins at once. It raises the value of working more, because additional liquid saving pays off precisely in the bad-benefit state where Social Security wealth is impaired (Proposition 1). And it raises the value of claiming early, because claiming converts part of a risky, non-tradable future claim into a current payment that the insolvency draw cannot touch (Proposition 2). The two responses are not competing explanations; they are two faces of the same self-insurance motive, and both strengthen as perceived insolvency risk rises. This is exactly the comovement the empirical design is built to detect, and the model shows it can arise from a single, transparent friction.

What the model takes as given, however, is the risk it is built around. Both p and δ are exogenous parameters: the household reacts to perceived insolvency, but its choices have no effect on actual solvency. This is where the model is silent on something quantitatively important, because working and claiming push the Social Security trust fund in *opposite* directions. Working longer keeps payroll-tax contributions flowing into the system and delays the date at which benefits are drawn, both of which slow the depletion of the trust fund. Claiming early does the reverse: it starts the benefit stream sooner and, by accompanying earlier labor-force exit, cuts off the contributions that would otherwise have continued. So the same insolvency fear that the model shows raising both margins generates a fiscal feedback the model cannot sign—one behavior shores up the fund while the other draws it down.

Which force dominates is not something a two-period model with fixed p and δ can resolve; it is a quantitative question. The net effect on solvency depends on the relative sizes of the work and early-claiming responses, on how much additional payroll revenue an extra year of work delivers, and on how much faster the fund is drawn down when claiming is pulled forward. Answering it requires endogenizing the trust-fund path and the contribution-and-benefit accounting that the simple model abstracts from, which is one of the tasks I take up in the full life-cycle model.

4. Life-Cycle Model

This section develops a structural life-cycle model in which a forward-looking household chooses consumption, labor supply, and the timing of Social Security claiming under uncertainty about health, wages, medical expenses, and employment. The model is annual.

The recursive problem is defined from age $t = 25$ through age 99, with certain death at age 100, but the quantitative model initializes the empirical cohort at age 55. The main computational cohort is born in 1941, so age 55 corresponds to calendar year 1996; the age-55 empirical distribution is pooled from the 1941–1945 HRS birth cohorts. In the forward overlapping-generations simulation, birth cohorts 1941–1965 enter the stochastic model at age 55 with the same empirical state distribution but different calendar-year Social Security belief paths. Ages 25–54 are treated as a deterministic pre-initialization block: labor is fixed at full-time work, survival is one, health is fixed, medical spending is zero, and perceived Social Security benefit risk is inactive. Thus the only active pre-55 household margin is the consumption-saving choice. These deterministic pre-55 years are still included in the trust-fund calendar ledger as payroll-tax inflows for cohorts that have not yet reached the stochastic initialization age. From age 55 onward the household faces the stochastic risks in the model and chooses consumption c_t, labor supply n_t, and, at most once, the age $q \in \{62, \ldots, 70\}$ at which to claim Social Security.

The household's problem is summarized by a vector of dynamic state variables together with a fixed type. The endogenous states carried forward are assets a_t, subject to the borrowing constraint $a_t \geq 0$; the Average Indexed Monthly Earnings (AIME) $\bar{e}_t$ that determines the benefit; the claiming age $q_{t-1} \in \{\emptyset, 62, \ldots, 70\}$, where $\emptyset$ denotes not yet claimed; and a small history state $\mathcal{J}_t$ that records the lagged labor-force information needed for re-entry costs. The stochastic states carried forward are health h_t, a discrete age-dependent Markov process; a persistent AR(1) wage shock ξ_t; and a persistent medical-expense shock κ_t. Unemployment $\iota_t \in \{0, 1\}$, the transitory medical shock ε_t, and the perceived benefit-risk draw introduced below are i.i.d. within-period draws—observed inside the period but not carried forward. The fixed heterogeneity, assigned before the household enters the stochastic part of the model and constant thereafter, is the type $\vartheta = (\theta, e)$, comprising education e and a permanent productivity type θ. The medical-expense process is estimated directly at this education-by-productivity type level rather than expanding the policy-function state space by Medicare eligibility or employer-insurance type. Thus $\mathcal{J}_t$ is a compact deterministic labor-history variable, not a separate insurance-type state. Collecting the dynamic states,

$$\boldsymbol{x}_t = (\underbrace{a_t, \bar{e}_t, q_{t-1}, \mathcal{J}_t}_{\text{endogenous}}\,;\underbrace{h_t, \xi_t, \kappa_t}_{\text{stochastic}}), \qquad \vartheta = (\theta, e),$$

with the value function written $V_t^{\vartheta}(\boldsymbol{x}_t)$. The production state space uses six fixed types, crossing two education groups and three permanent productivity groups. The first production grid uses 25 asset nodes and 10 AIME nodes. The carried shocks are discretized with three wage nodes and three persistent medical-expense nodes; the within-period transitory medical shock uses three quadrature nodes and the benefit-risk draw uses two outcomes. Because insolvency beliefs are calendar-year objects, I solve the dynamic program separately for each birth year from 1941 to 1965, remapping the age profile of perceived benefit risk through $t = c + a$ for each cohort c. Production runs use compact temporary policy storage by birth year and fixed type for the forward simulation pass, rather than permanent full policy-function exports.

Benefits enter through a perception channel that is central to the mechanism. Let p_t denote the annualized perceived probability that scheduled benefits are at risk in calendar year t. In the household problem this probability is period specific, not an absorbing reform hazard. Thus the perceived benefit factor before the period's benefit-risk draw is

$$p_t\,[1-\delta]\,b + (1-p_t)\,b = [1-p_t\,\delta]\,b,$$

where δ is the constant cut severity.

4.1. Environment

4.1.1. Preferences

Each agent gets a flow utility from consumption c_t and leisure ℓ_t through a Cobb-Douglas consumption-leisure composite,

$$u(c_t, \ell_t) = \frac{(c_t^{\alpha}\ell_t^{1-\alpha})^{1-\gamma}}{1-\gamma},$$

where $\gamma > 0$ is the risk aversion and $\alpha \in (0,1)$ is the consumption share. Labor supply is denoted as working hours n_t.

The total amount of leisure is determined by

$$\ell_t = \overline{\ell} - n_t - \phi_{n,t}\mathbf{1}\{n_t \neq 0\} - \phi_{h,t}h_t - \phi_{RE}\mathrm{RE}_t, \tag{1}$$

where $\overline{\ell}$ is total annual time endowment and $\mathbf{1}$ is an indicator function. The participation fixed cost $\phi_{n,t}$, paid whenever $n_t \neq 0$, generates a non-convexity that induces households to exit the labor market (Rogerson and Wallenius 2013). The health-dependent fixed cost $\phi_{h,t}$ scales with health h_t, since the leisure available for work differs by health type. Note that both fixed costs are also time-varying. This is to capture the increasing adverse effect of aging on labor supply. The last term ϕ_{RE} is to capture the scarring effect of unemployment in labor market as it becomes much harder to find an alternative job when unemployed. $RE_t = \mathbf{1}\{n_{t-1} = 0,\ n_t > 0\}$ is the re-entry indicator, as in French and Jones (2011); it depends on the past only through the binary participation indicator $E_t^- = \mathbf{1}\{n_{t-1} > 0\}$, so the level of lagged hours need not be carried as a state.

Following De Nardi (2004) and Lockwood (2018), I assume a warm-glow bequest function

$$\nu_\vartheta(a) = \theta_b \frac{(a + \underline{a})^{(1-\gamma)\eta_\vartheta}}{1-\gamma},$$

where θ_b is the common bequest intensity, $\underline{a}$ shifts the bequest motive so that bequests behave as a luxury good, and η_ϑ is a fixed-type multiplier on the curvature of the bequest function. The multiplier changes the curvature of late-life asset demand across education-by-productivity types without mechanically scaling the level of the bequest motive. When $\eta_\vartheta = 1$, the bequest term reduces to the standard CRRA warm-glow specification. This keeps senior households from spending down saving too fast, and it lowers the marginal value of annuitized income which feeds directly into the claiming decision. In other words, a strong bequest motive makes the Social Security annuity less valuable and tilts toward early claiming.

4.1.2. Health and Mortality

Health and mortality are first-order determinants of late-life labor supply and claiming, and they enter the model as exogenous, age-dependent risks. Health h_t evolves as a discrete Markov process whose transition probabilities depend on age and on the permanent type, so that the likelihood of moving into bad health rises with age and is higher for low-education, low-productivity households. Health is not a passive state but shapes behavior through three channels. First, it lowers the effective time available for work through the health-dependent fixed cost $\phi_{h,t}$, making continued employment more costly as health deteriorates. Second, it raises both the level and the dispersion of medical expenses in the process described below, strengthening the precautionary motive when earnings capacity is falling. Third, because health predicts mortality, it governs the horizon over which the household values future consumption, saving, and Social Security benefits.

Survival from age t to $t+1$ occurs with probability $\pi_t(\theta, e)$, which depends on age and on the permanent type (θ, e) after the model is initialized at age 55. For ages below 55 in the computational recursion, I set $\pi_t = 1$ and use identity health transitions. This dependence captures the well-documented socioeconomic gradient in longevity: higher-educated, higher-productivity households live longer. The heterogeneity matters for claiming because Social Security is a life annuity whose actuarial adjustment for delay is actuarially fair at average mortality. A household that expects to be long-lived earns a higher implicit return from postponing claiming and from the additional benefit accrual that continued work brings, whereas a household facing high mortality risk values the early, certain receipt of benefits and discounts the future more steeply. Differential mortality generates dispersion in optimal claiming ages and in the labor-supply response.

Each household discounts the future at rate β, weights it by the probability of remaining alive, and upon death derives warm-glow utility from bequeathed assets. In the recursive problem the household maximizes

$$\mathbb{E} \sum_{t=25}^{99} \beta^{t-25} \left(\prod_{k=25}^{t-1} \pi_k \right) \left[u(c_t, \ell_t) + \beta(1-\pi_t)\nu_{\vartheta}(a_{t+1}) \right],$$

where $\prod_{k=25}^{t-1} \pi_k$ is the probability of surviving to age t and $\nu_{\vartheta}(a_{t+1})$ is the warm-glow value of the asset left with death probability $1-\pi_t$. Mortality thus enters both as a survival weight on future utility flows and as the trigger for the bequest motive, linking longevity heterogeneity to the speed of asset decumulation and to the value households place on the Social Security annuity.

4.1.3. Wages

Unemployment is an i.i.d. shock that strikes with probability π^{ι} each period; it is observed at the start of the period but is not carried forward as a state. The effect of unemployment, $\iota_t = 1$, is a lower labor productivity and hence a lower hourly wage. Persistence in labor-market disadvantage operates not through the wage—which $\chi(\iota_t)$ depresses only contemporaneously—but through the re-entry cost in the leisure technology, which penalizes returning to work after a spell of nonemployment (Bairoliya and McKiernan 2023).

$$\omega_t = \chi(\iota_t) \cdot \exp(w_t(\theta, e) + \xi_t),$$

where $\chi(\iota_t)$ is the unemployment productivity penalty ($\chi(0) = 1$, $\chi(1) < 1$), $w_t(\theta, e)$ is the deterministic age–type wage profile, and the persistent wage shock follows

$$\xi_t = \rho \xi_{t-1} + \zeta_t^w, \quad \zeta^w \sim \mathcal{N}(0, \sigma_w^2).$$

4.1.4. Medical Expenditure

Medical expenses are the central uninsurable risk of late life. They rise steeply with age, are highly skewed, and are only partially covered by public and private insurance. Because these expenses are large, persistent, and concentrated near the end of life, they generate a strong precautionary saving motive and discourage households from running down their wealth, even at advanced ages. French, Jones, and McGee (2023) and De Nardi et al. (2025) explain why the elderly draw down assets so slowly, and they show that the balance between self-insurance and bequest motives is not uniform but varies systematically across households. The relative weight of these two motives depends on permanent income, marital status, and age. In the first full model run, I use the HRS out-of-pocket measure augmented with the French-Jones Medicaid payment imputation and let medical expenses vary by age, health, labor-force participation, and permanent type. I do not split the production medical process by Medicare eligibility or employer health-insurance type, because those distinctions are not the target source of variation for the insolvency-risk experiment.

I model log medical expenses as the sum of a deterministic and a stochastic component,

$$\ln(M_t) = m_t(h_t, \mathbf{1}\{n_t > 0\}, \theta, e) + \sigma_t(\theta, e) \cdot \psi_t,$$

where h_t is health, $\mathbf{1}\{n_t > 0\}$ is labor-force participation, and (θ, e) is the permanent productivity-by-education type; age enters through the time subscript t. The conditional mean $m_t(\cdot)$ captures how the level of medical spending shifts with age, health status, labor-force participation, and type, while the volatility term $\sigma_t(\cdot)$ allows the dispersion of medical-expense risk to vary over age and type. Letting both the mean and the variance vary across these states is what reproduces the age profile and the wide cross-sectional spread of medical expenses.

The individual-specific shock ψ_t combines a persistent and a transitory component,

$$\psi_t = \kappa_t + \varepsilon_t, \qquad \kappa_t = \rho_m \kappa_{t-1} + \eta_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2),$$

where κ_t is a persistent AR(1) component with auto-correlation ρ_m and ε_t is an i.i.d. transitory shock. An adverse draw to κ_t raises the conditional expectation of future medical expenses and is not immediately reversed, whereas ε_t captures idiosyncratic shocks. It is the persistence in κ_t that makes medical expenses a driver of saving and of the decision to delay claiming and retirement.

4.1.5. Health Insurance

Health insurance affects late-life labor supply because the value of continued employment depends on whether employer coverage is retained after retirement. Following French and Jones (2011), I distinguish three employer-related coverage assignments: retiree coverage, tied coverage, and no employer coverage. Retiree coverage continues after labor-force exit, tied coverage is available only while working before Medicare eligibility, and households without employer coverage face the no-coverage medical-spending environment.

In the model employer-insurance assignment is not part of the solved fixed type. I estimate the age-55 shares of retiree, tied, and no-coverage households within each fixed-type from HRS data, and use those shares to average the medical-expense environment faced by that type. Medicare covers all households from age 65. This keeps the health-insurance channel in the model without introducing a separate insurance-type state or transition in the paper notation.

4.2. Social Security and its Trust Fund

In a frictionless world Social Security would be the safe, annuitized retirement wealth, a government-backed annuity that insures longevity and inflation risk. Its payout, however, is partly financed by a trust fund whose solvency households increasingly doubt. As the perceived probability of depletion rises, households stop treating the benefit as a safe asset and price it as risky. In other words, a claim whose realized value is uncertain and which cannot be sold or hedged by the law. They can respond only by self-insuring, such as accumulating private wealth and claiming earlier. Households cannot rebalance their portfolio and this constraint carries the welfare cost.

The novel feature of this paper is that the trust fund is endogenous. Revenue and outlays aggregate the same labor supply and claiming decisions that perceived risk distorts, so beliefs about insolvency feed back into the fund's dynamic path and the benefit cut on top of the demographic forces driving its secular decline. In addition, the perceived risk rises over time, so the distortion deepens across cohorts or generations. This implies that a younger generation is exposed to higher perceived probability of Social Security insolvency which distorts their behavior more than an older generation.

A household claiming at age q is entitled to a scheduled benefit equal to its Primary Insurance Amount, adjusted for the claiming age and net of the retirement test,

$$b_t(q) = \max\left\{ \Gamma(q)\,\mathrm{PIA}(\bar{e}_t) - \Upsilon_t,\ 0 \right\}, \tag{2}$$

where $\Gamma(q)$ is the actuarial adjustment for early or delayed claiming and Υ_t the amount withheld under the earnings test if claimed before full retirement age. This is what the current law promises and whether it is paid full depends on the fund. This is the part which I depart from Bairoliya and McKiernan (2023) since they model misperception of the adjustment $\Gamma(q)$ to account for early claiming. The implication of their approach is that an idiosyncratic and static distortion that resolves at the claiming, after which the annuity is treated as a safe asset. I hold $\Gamma(q)$ correctly perceived but let households doubt that scheduled benefits will be paid at all, an expectation that does not resolve at claiming. The schedule benefit remains risky throughout retirement.

I model this doubt as period-specific perceived benefit risk. Let $\zeta_t \in \{0, 1\}$ denote the period-t benefit-risk draw, where $\zeta_t = 1$ is the adverse draw in which scheduled benefits are perceived to be cut by δ. The empirical series p_t constructed in Appendix B gives

$$\Pr(\zeta_t = 1) = p_t,$$

and the perceived payment is

$$s_t(q, \zeta_t) = \begin{cases} b_t(q), & \zeta_t = 0, \\ [1-\delta]\, b_t(q), & \zeta_t = 1. \end{cases} \tag{3}$$

The draw is not an absorbing reform state and is not carried into the next period. Instead, households update the perceived risk each calendar year using the empirical path of p_t. Thus perceived insolvency risk rises because the measured belief series rises over time, not because one-year hazards compound mechanically. Since the household chooses labor supply and claiming before ζ_t is realized, the perceived payment has mean

$$\mathrm{E}_t[s_t] = [1 - p_t\,\delta]\; b_t(q), \tag{4}$$

and variance

$$\mathrm{Var}_t(s_t) = p_t(1 - p_t)\delta^2\; b_t(q)^2. \tag{5}$$

The variance is zero only when $p_t = 0$ or $p_t = 1$ and it rises over the relevant range as perceived risk increases. The reclassification of Social Security from a safe to a risky asset is this variance term becoming positive, and rising perceived insolvency is its upward trend. A risk-averse household responds to both moments. The level in (4) lowers the benefit, while the variance in (5) erodes the annuity's insurance value and drives households into inefficient private self-insurance because the claim is non-tradeable.

The trust fund. The benefit households actually receive is fixed by the fund, which aggregates the population's decisions through the OASI accounting identity. Let $\Phi_{t,j}(\boldsymbol{x})$ be the measure of age-j households in state $\boldsymbol{x}$ at calendar time t. Payroll-tax revenue is covered earnings, taxed at the OASI rate τ_{ss} up to the taxable maximum $\bar{y}_t^{ss}$,

$$T_t = \tau_{ss} \sum_{j=25}^{99} \int \min\bigl(\omega_j(\boldsymbol{x})\; n_j^*(\boldsymbol{x}),\;\; \bar{y}_t^{ss}\bigr)\; d\Phi_{t,j}(\boldsymbol{x}), \tag{6}$$

and scheduled outlays are benefits to claimants,

$$G_t = \sum_{j=62}^{99} \int \max\bigl\{b_{t,j}(\boldsymbol{x}) - \Upsilon_{t,j},\; 0\bigr\}\,\mathbf{1}\{\text{claimed}\}\; d\Phi_{t,j}(\boldsymbol{x}), \tag{7}$$

with $b_{t,j}(\boldsymbol{x})$ the scheduled benefit (2) evaluated at the household's claim age and AIME. Because labor supply and claiming decision are optimal responses to perceived insolvency risk, both flows are endogenous to beliefs. The fund's trajectory is shaped by the behavior that doubting the fund induces. The fund accumulates the gap between revenue and realized outlays at the special-issue rate r_t^F under a no-borrowing rule. Scheduled benefits are paid in full while reserves and current revenue suffice, and otherwise are scaled to the payable level,

$$\text{(8)} \qquad \chi_t = \min\left[1,\ \frac{(1+r_t^F)F_t + T_t}{G_t}\right], \qquad F_{t+1} = \max\left\{0,\ (1+r_t^F)F_t + T_t - G_t\right\},$$

so realized outlays equal $\chi_t G_t$ and reserves never turn negative. The payable factor χ_t equals one while the fund is solvent and falls below one at depletion, after which the system is pay-as-you-go. The depletion year is $t^0 = \min\{t : \chi_t < 1\}$. This annual proportional rule is a tractable statutory fact that scheduled benefits cannot be paid in full after reserve exhaustion absent legislation. A claimant therefore receives the realized benefit $\chi_t\, b_t(q)$, the scheduled benefit scaled by what the fund can finance.

4.3. Recursive Formulation

The problem is solved separately for each fixed type $\vartheta = (\theta, e)$, whose index I suppress. Employer-insurance assignment is averaged within ϑ. A household enters age t with the predetermined state

$$\boldsymbol{x}_t = \big(\underbrace{a_t, \bar{e}_t, q_{t-1}, \mathcal{J}_t}_{\text{endogenous}};\ \underbrace{h_t, \xi_t, \kappa_t}_{\text{stochastic}}\big),$$

where q_{t-1} is the claiming status carried in from period $t-1$, in force at the start of the period. The order of moves within the period is the crux of the mechanism,

$$\text{(9)} \qquad \boldsymbol{x}_t \longrightarrow \iota_t \longrightarrow (n_t, d_t) \longrightarrow (\varepsilon_t, \zeta_t) \longrightarrow (c_t, a_{t+1}) \longrightarrow \boldsymbol{x}_{t+1}.$$

The period unfolds in two stages. First, before any of the period's risks resolve, the household observes only the unemployment draw ι_t—which pins down its wage ω_t—and on that basis commits its labor supply n_t and its claiming decision d_t. Because these choices are locked in before the medical shock ε_t and the benefit-risk draw ζ_t arrive, they cannot react to those shocks directly: the household factors them in expectation, using the calendar-year perceived probability p_t. Only then are (ε_t, ζ_t) realized, and the household makes its final choice of consumption and saving. This ordering matters for two reasons. Medical and perceived benefit risk reach the work and claiming margins only through the continuation value and the bequest, never as information in hand when those choices are made, so labor supply and claiming are precautionary. Consumption, chosen after the shocks arrive, is the flexible buffer that absorbs them and keeps the budget balanced state by state. Deferring it is also what keeps the consumption floor coherent: were c_t fixed in advance, a large enough medical expense M_t could force next-period assets below zero, $a_{t+1} < 0$.

The claiming decision $d_t \in \{0, 1\}$ ($d_t = 1$ meaning an as-yet-unclaimed household claims at t) updates the status through $q_t = Q_t(q_{t-1}, d_t)$, equal to q_{t-1} if already claimed, t if the household claims now, and $\emptyset$ otherwise. Claiming is irreversible, so benefits begin in the period of claiming. The feasible set for (n_t, d_t) given the carried-in status is

$$\mathcal{D}_t(q_{t-1}) = \begin{cases} \{0\}, & t < 62 \text{ or } q_{t-1} \neq \emptyset, \\ \{0,1\}, & 62 \leq t < 70,\ q_{t-1} = \emptyset, \\ \{1\}, & t = 70,\ q_{t-1} = \emptyset. \end{cases} \tag{10}$$

With labor earnings $y_t = \omega_t n_t$, the scheduled benefit evaluated is

$$b_t(q_t, \bar{e}_t, n_t) = \mathbf{1}\{q_t \neq \emptyset\} \max\big\{ \Gamma(q_t)\, \mathrm{PIA}(\bar{e}_t) - \Upsilon_t,\ 0 \big\},$$

and the perceived payment is $s_t(q_t, \zeta_t)$ in (3): the household keeps the full b_t when $\zeta_t = 0$ and receives $[1-\delta] b_t$ when $\zeta_t = 1$. The household chooses labor and claiming before this period's draw ζ_t is known, so it values the benefit at its expectation, $\mathbb{E}_t[s_t] = [1 - \delta\, p_t]\, b_t$. This expected benefit is what drives the household's choices. It differs from what the program actually pays on the true fund path, $s_t^A = \chi_t\, b_t$, where χ_t of (8) is the share of scheduled benefits the trust fund can finance.

Budget Constraint. With gross return $R_t \equiv 1 + r_t$, household taxes $\mathcal{T}_t$, and means-tested transfers tr_t, next-period assets are the residual of the end-of-period draws (ε_t, ζ_t),

$$\begin{aligned} a_{t+1}(\varepsilon_t, \zeta_t) = R_t a_t + &\ y_t + s_t(q_t, \zeta_t) + \mathrm{tr}_t(\varepsilon_t, \zeta_t) \\ &- \mathcal{T}_t\big(r_t a_t,\ y_t,\ s_t(q_t, \zeta_t)\big) - c_t - M_t(\varepsilon_t), \\ &a_{t+1} \geq 0,\ c_t \geq \underline{c},\ \ell_t > 0, \end{aligned} \tag{11}$$

where s_t is the perceived payment in the household's problem, and the tax $\mathcal{T}_t$ is levied on asset income $r_t a_t$ and covered earnings. The notation $M_t(\varepsilon_t)$ keeps the budget compact: conditional on $(\boldsymbol{x}_t, n_t, \vartheta)$, the transitory draw ε_t is the remaining random argument, while dependence on health, persistent medical risk κ_t, the history state, and type is implicit. Means-tested transfers implement the consumption floor $\underline{c}$ in the manner of Hubbard, Skinner, and Zeldes (1994, 1995), covering any shortfall of after-tax cash on hand—gross assets $R_t a_t$, labor, and Social Security, net of out-of-pocket medical spending—below $\underline{c}$,

$$\begin{aligned}\mathrm{tr}_t(\varepsilon_t,\zeta_t) = \max\Big\{0,\ \underline{c} + M_t(\varepsilon_t) \\ - \big[\, R_t a_t + y_t + s_t(q_t,\zeta_t) - \mathcal{T}_t\big(r_t a_t,\ y_t,\ s_t(q_t,\zeta_t)\big)\,\big]\Big\}.\end{aligned} \tag{12}$$

Because the transfer nets medical spending and the means test is on gross cash on hand, it guarantees post-medical consumption of at least $\underline{c}$. With consumption chosen after (ε_t,ζ_t), an active transfer forces $c_t = \underline{c},\ a_{t+1} = 0$, so households exhaust private wealth before drawing support and cannot use transfers to fund saving.

The household's problem differs across the life cycle only in the feasible set for the pre-shock controls (n_t, d_t). Combining the claiming window (10) with a mandatory labor-force exit at age 81, the control set $\mathcal{A}_t(\boldsymbol{x}_t)$ partitions the life cycle into four phases,

$$\mathcal{A}_t(\boldsymbol{x}_t) = \begin{cases} \{(n,0) : n \geq 0\}, & 25 \leq t < 62 \ \text{(working)}, \\ \{(n,d) : n \geq 0,\ d \in \mathcal{D}_t(q_{t-1})\}, & 62 \leq t \leq 70 \ \text{(claiming)}, \\ \{(n,0) : n \geq 0\}, & 71 \leq t \leq 80 \ \text{(post-claiming)}, \\ \{(0,0)\}, & 81 \leq t \leq 99 \ \text{(retired)}. \end{cases} \tag{13}$$

Before 62 claiming is infeasible and the household chooses labor alone; from 62 to 70 it additionally chooses whether to claim, with the claim forced at 70 if still deferred (10); from 71 to 80 claiming is settled and only labor remains; and from 81 on, labor is fixed at zero, leaving only the post-shock consumption–saving choice.

Recursive representation. Denote $\mathbb{E}_t^z[\cdot]$ for the expectation over $\boldsymbol{z}_{t+1} = (h_{t+1}, \xi_{t+1}, \kappa_{t+1})$ under F_t^z. The value function has a nested form: an outer choice of (n_t, d_t) taken before the draws, wrapped around an inner choice of (c_t, a_{t+1}) taken after them. The outer problem is

$$V_t(\boldsymbol{x}_t) = \mathbb{E}_t^\iota\left[\max_{(n_t,d_t)\in\mathcal{A}_t(\boldsymbol{x}_t)} \sum_{\zeta_t\in\{0,1\}} \Pr_t(\zeta_t) \int W_t\big(\boldsymbol{x}_t, \iota_t, n_t, d_t, \zeta_t, \varepsilon_t\big)\, dF_t^\varepsilon(\varepsilon_t)\right], \tag{14}$$

and the inner, post-shock problem is

$$W_t\big(\boldsymbol{x}_t, \iota_t, n_t, d_t, \zeta_t, \varepsilon_t\big) = \max_{\substack{c_t \geq \underline{c} \\ a_{t+1} \geq 0}} \Big\{ u(c_t, \ell_t) + \beta\big[\, \pi_t\, \mathbb{E}_t^{z} V_{t+1}(\boldsymbol{x}_{t+1}) + (1-\pi_t)\, \nu_\vartheta(a_{t+1}) \,\big] \Big\}, \tag{15}$$

subject to the budget constraint (11), with leisure ℓ_t given by (1). The maximization over (n_t, d_t) sits outside the expectation over (ε_t, ζ_t) and the maximization over (c_t, a_{t+1}) inside it. Survival $\pi_t(\theta, e)$ weights the continuation value, and the warm-glow bequest $\nu_\vartheta(a_{t+1})$ is collected with probability $1 - \pi_t$.

Because the phases (13) enter the recursion only through the feasible set $\mathcal{A}_t$, the inner problem (15) is the same at every age: once the shocks land, the household always solves the same consumption–saving problem. What changes with age is only the outer menu of (n_t, d_t), and it shrinks as the household gets older—labor alone before claiming opens, labor together with the claim decision during the claiming window, and labor alone again once the claim is settled. In full retirement the menu collapses to a single point, $(n_t, d_t) = (0, 0)$, so the outer maximization drops out and the value function is just the expected continuation,

$$V_t(\boldsymbol{x}_t) = \mathbb{E}_t^{\iota}\left[\sum_{\zeta_t \in \{0,1\}} \Pr_t(\zeta_t) \int W_t\big(\boldsymbol{x}_t, \iota_t, 0, 0, \zeta_t, \varepsilon_t\big)\, dF_t^{\varepsilon}(\varepsilon_t) \right],$$

a pure consumption–saving problem under medical and perceived benefit risk. At the final age survival is zero, $\pi_{99} = 0$, so the continuation in (15) reduces to the bequest $\nu_\vartheta(a_{100})$ alone.

Conditional on survival, the next-period state stacks the post-decision claiming status $q_t = Q_t(q_{t-1}, d_t)$, the updated history state $\mathcal{J}_{t+1}$, the AIME accrual $\bar{e}_{t+1}$, and the next-period stochastic states $\boldsymbol{z}_{t+1}$. The history state records only the information from current choices that matters next period: recent work for re-entry costs and, before Medicare, retained retiree coverage for medical spending. Its update is deterministic, so no separate stochastic insurance-state transition is introduced:

$$\boldsymbol{x}_{t+1} = \big(a_{t+1}, \bar{e}_{t+1}, q_t, \mathcal{J}_{t+1}; h_{t+1}, \xi_{t+1}, \kappa_{t+1}\big).$$

The AIME accrues from current covered earnings $y_t = \omega_t n_t$ through the running top-35 average

$$\bar{e}_{t+1} = \min\Big[\, \bar{e}_t + \max\Big(0,\ \tfrac{1}{35}\big[\min(y_t,\ \bar{y}_t^{ss}) - \bar{e}_t\big]\Big),\ \bar{y}_t^{ss}\,\Big], \tag{16}$$

so a year of covered earnings above the running average raises $\bar{e}_{t+1}$ and a year below leaves it unchanged; the indexation and the 35-year window are detailed in Appendix A.1.

5. Estimation

5.1. First Stage Estimation

5.1.1. Health and Survival

Health and mortality are first-stage objects: I estimate them outside the model and hold them fixed during preference estimation. Health h_t is a three-state Markov process—good, fair, and bad—whose transition probabilities depend on age and on the permanent type (θ, e), and survival $\pi_t(\theta, e)$ depends on the same states. I estimate the health transitions from the Medical Expenditure Panel Survey (MEPS) and the survival probabilities from MEPS at younger ages and the HRS from age 50 on. These objects enter the model directly: health shifts the fixed cost of work and the medical-expense process, while survival weights continuation utility and triggers the bequest.

The economically important feature is the gradient. Low-education, low-productivity households deteriorate into bad health faster and die earlier, and bad health today raises both future health deterioration and mortality (Figures 5 and 6). This dispersion in horizons is what makes longevity heterogeneity matter for behavior: long-lived households gain more from delaying claiming and from saving, whereas high-mortality households value early, certain benefit receipt and discount the future more steeply. Appendix C.3 reports the data construction, the ordered-probit specifications, and the age ranges over which each source is used.

5.1.2. Social Security Trust Fund

I track the OASI trust fund in normalized units—the reserve ratio F_t/G_t and the revenue ratio T_t/G_t—and apply the same payable-benefit law used in the model. The fund starts in 1996 at the actual OASI trust-fund ratio, $F_{1996}/G_{1996} = 1.93$. I use OASI rather than the combined OASDI fund because Disability Insurance benefits are not part of the household's benefit in the model.

Two adjustments align the 1996 starting point with the real program. First, the outlay denominator G_{1996} must also count benefits paid to people already retired when the simulation begins not only the model cohorts. I anchor this legacy cost to the official SSA OASI total outlay for 1996, converted into the model's dollar units and net of any 1996 benefits already paid by the simulated cohorts. Second, I initialize the legacy claimant stock from SSA current-payment OASI beneficiary counts, weighted by retired-worker age and sex from the Social Security Annual Statistical Supplement, and age it forward with the model's survival schedule, so legacy outlays fall over time as these beneficiaries die.

The fund is an aggregate object, not a single cohort's account: revenue T_t and outlays G_t are calendar-year flows summed over every cohort alive in year t. The forward simulation therefore tracks all birth cohorts from 1941 to 1965, sizing each by Census population estimates at its age-55 entry year and adding the deterministic payroll taxes it pays before age 55, the point at which it enters the stochastic part of the model. Any single-cohort fund path is only a diagnostic.

I validate the model against the official benchmark that OASI reserves are depleted in 2032 (SSA Trustees). The baseline feeds in the measured perceived benefit-risk probabilities through 2026 and then holds them at their 2026 level. The counterfactual instead freezes perceived risk at its 1996 level from 1996 onward, re-solves the cohort policy functions under that belief path, and compares the resulting endogenous depletion year with the 2032 benchmark.

The model contains a feedback loop between beliefs and outcomes. Households act on the benefit they expect to receive, treating it as riskier when the perceived insolvency probability p_t is higher. Their work, saving, and claiming decisions aggregate into payroll revenue T_t and outlays G_t, which set the payable factor χ_t and the depletion year t^0—and χ_t is exactly what fixes the benefit households ultimately collect. Beliefs shape behavior, behavior shapes the fund, and the fund shapes the very benefits the beliefs were about.

What matters for welfare is the gap between fear and reality: the perceived insolvency risk p_t against the realized shortfall $1 - \chi_t$. When households are more pessimistic than the fund's actual path warrants, they self-insure by saving more and claiming earlier, because a Social Security claim cannot be sold or hedged. This response feeds back into the fund with an ambiguous sign: earlier claiming pulls outlays forward, while extra work and saving raise payroll revenue. Whether pessimism ends up confirming or undoing itself is therefore a quantitative question—which is why this paper treats the fund as endogenous rather than as a fixed external projection.

5.1.3. Taxes

The household tax function $\mathcal{T}_t\big(r_t a_t,\ y_t,\ s_t\big)$ that enters the budget constraint (11) has two components. First, the OASI payroll tax is levied at rate τ_{ss} on covered labor earnings up to the taxable maximum $\bar{y}_t^{ss}$, $\tau_{ss}\min(y_t,\ \bar{y}_t^{ss})$. This is the same flow that finances the trust fund through revenue (6). Second, a smooth progressive income tax applies to the taxable base

$$z_t = y_t + r_t a_t + \chi^{ss} s_t,$$

which comprises labor earnings, asset income, and the taxable portion of Social Security benefits. Following the tax-progressivity specification of Heathcote, Storesletten, and Violante (2017), after-tax income is a log-linear function of pre-tax income,

$$z_t - \mathcal{T}_t^{\text{inc}}(z_t) = \bar{z}^T\,\lambda_t^T \left(\frac{z_t}{\bar{z}^T}\right)^{1-\tau_t^T},$$

where λ_t^T controls the average level of the non-OASI income tax at the reference income $\bar{z}^T$, and τ_t^T controls progressivity. In the computation I floor $\mathcal{T}_t^{\text{inc}}(z_t)$ at zero because the means-tested consumption floor below is the model's explicit transfer mechanism. Thus

$$\mathcal{T}_t\big(r_t a_t,\ y_t,\ s_t\big) = \tau_{ss}\ \min(y_t,\ \bar{y}_t^{ss})\ +\ \max\Big\{\mathcal{T}_t^{\text{inc}}(z_t), 0\Big\}, \tag{17}$$

where $\chi^{ss} \in [0, 1]$ is the share of benefits included in taxable income. Payroll taxes are not collected on benefits or asset income, and earnings above $\bar{y}_t^{ss}$ are neither taxed for OASI nor credited toward the AIME in (16).

5.2. Second Stage Estimation

The second stage estimates the preference parameters for the main 1941–1945 male household-head birth cohorts, holding the first-stage environment fixed. These are the cohorts that reach the age-55 stochastic initialization in the mid-1990s and whose retirement-age behavior the model is asked to reproduce. Estimation is by the simulated method of moments (SMM). For a vector of preference parameter Θ, I simulate the model, form the same moments in the simulated and HRS samples, and minimize their weighted distance. SMM and the closely related method of simulated moments are standard tools for estimating life-cycle models whose moments have no closed form (Pakes and Pollard 1989; Duffie and Singleton 1993), and a large structural literature on consumption, saving, and retirement uses them in exactly this way (Gourinchas and Parker 2002; Cagetti 2003;

French 2005; French and Jones 2011).

5.2.1. Data

The second-stage calibration uses three empirical objects from the HRS. First, I target employment rates for the 1941–1945 male household-head cohorts in four retirement-age bins: 62–65, 66–69, 70–73, and 74 and older. These moments discipline the discount factor and the fixed cost of work because they summarize the slope of retirement-age labor supply after the initial age-55 state distribution has been fixed.

Second, I target the late-age distribution of saving for male household-head observations at ages 85 and older. The asset is liquid financial assets excluding IRA balances, net of debt when the underlying components are available. Monetary quantities are expressed in thousands of 2020 dollars. I match empirical moments: the mean, 25th percentile, and 75th percentile of late-age saving, both overall and separately by fixed type. These moments identify the common bequest intensity and shift together with the fixed-type curvature multipliers in $\nu_{\vartheta}(a)$.

Third, I use the claiming-age distribution as a validation and selection restriction for the final calibration. The claiming target is constructed from the same 1941–1945 male household-head cohorts, restricting to respondents observed unclaimed before age 62 and later observed claiming, and excluding respondents with positive SSI or SSDI income before or at the first observed claim. These moments are informative because both patience and the bequest motive change the value of the Social Security annuity and therefore the incentive to claim early.

5.2.2. Estimation and Identification

Let m denote the vector of targeted employment moments, late-age saving moments, and claiming-age validation moments. For parameter vector

$$\Theta = (\beta, \phi_{n,0}, \phi_{n,1}, \theta_b, \underline{a}, \eta_1, \ldots, \eta_6)\,,$$

the simulated method of moments (SMM) objective minimizes the weighted distance between the HRS targets and their simulated analogs,

$$Q(\Theta) = (m(\Theta) - m^{data})'\, W\, (m(\Theta) - m^{data}),$$

where $m = (m_E', m_A', m_C')'$ stacks the employment, saving, and claiming moments and $W = \text{diag}(W_E, W_A, W_C)$ is the block-diagonal weight matrix. The reported calibration uses diagonal weights and normalizes the wealth component by the current asset cap. The claiming block puts weight on the cumulative claiming-age distribution and the age-62 claiming share, because the original employment-and-wealth calibration overstated immediate claiming at age 62.

6. Model Fit

6.1. Targeted Moments

The second-stage calibration targets retirement-age employment and the late-age saving distribution. The employment target uses the RAND HRS 1941–1945 male household-head proxy sample in four age bins. Let $S_{B\tau}$ denote the HRS person-wave observations in age bin B and fixed type τ. The empirical moment is the weighted employment rate in that cell:

$$m_{B,\tau} = \frac{\sum_{(i,t)\in S_{B\tau}} \omega_{it} e_{it}}{\sum_{(i,t)\in S_{B\tau}} \omega_{it}},$$

where e_{it} is the employment indicator and ω_{it} is the HRS analysis weight. Table 1 (Panel A) reports the overall employment fit. The same age-bin moments by fixed type enter the calibration objective.

The late-age saving target uses male household-head proxy observations at ages 85 and older. The asset concept is liquid financial assets excluding IRA balances, matching the model asset state. Table 1 (Panel B) reports the overall fit to the mean and interquartile range.

The targeted moments show where the calibrated model is strongest and where it remains disciplined by the data. It preserves the declining retirement-age employment profile and matches employment at ages 66–69 closely. It also keeps the center of the late-age saving distribution in the right range: the model mean is about $25,000 above the capped HRS target and the upper quartile is within about $33,000.

The fit is less successful at the margins of the lifecycle distribution. Employment is too high at ages 62–65 but then falls too sharply after age 70, so the model understates work at ages 70–73 and 74 and older. The saving distribution is also too compressed near the bottom, putting too much mass away from near-zero liquid assets. Thus the calibration is useful for the average retirement-age profile and the level of late-life saving,

TABLE 1. Targeted moments and calibrated model fit

	HRS	Model	Gap
Panel A. Retirement-age employment			
62–65	0.591	0.683	+0.092
66–69	0.448	0.458	+0.011
70–73	0.340	0.228	-0.112
74+	0.239	0.087	-0.151
Panel B. Late-age saving distribution			
Mean	126.6	152.0	+25.4
25th percentile	2.1	49.2	+47.2
75th percentile	275.9	243.1	-32.8

Note. Panel A reports retirement-age employment, the weighted worked-for-pay rate, by age bin. The RMSE across the four employment bins is 0.105. Panel B reports the late-age liquid-saving distribution in thousands of 2020 dollars. The overall wealth RMSE is 36.3, and the fixed-type-weighted wealth RMSE across type-specific mean, 25th-percentile, and 75th-percentile moments is 130.3.

while heterogeneity in late-life resources and very late employment should be interpreted more cautiously.

6.2. Untargeted Moments

I next evaluate the claiming-age distribution as an untargeted moment. The final reported calibration is chosen to preserve the targeted employment and late-age saving fit. Table 2 compares the calibrated model with the HRS claiming-age target.

TABLE 2. Claiming-age moments

Moment	HRS	Model	Gap
Mean claiming age	64.17	64.29	+0.11
Age-62 claiming share	0.408	0.420	+0.012
Cumulative claimed by 65	0.763	0.706	-0.057
Age-70 claiming share	0.038	0.012	-0.026

Note. The HRS target is constructed from respondents observed unclaimed before age 62 and later observed claiming, excluding respondents with positive SSI or SSDI income before or at the first observed claim. The model CDF RMSE against the cumulative claiming-age target is 0.055, and the density RMSE is 0.053.

Table 3 reports the corresponding age-by-age claiming distribution. The selected calibration fixes the main problem in the employment-and-wealth-only fit: the age-62 claiming share is no longer too high, and the mean claiming age is within 0.11 years of

TABLE 3. Claiming-age distribution

Age	HRS share	Model share	Gap	HRS cumulative	Model cumulative	Gap
62	0.408	0.420	+0.012	0.408	0.420	+0.012
63	0.100	0.012	-0.088	0.509	0.432	-0.077
64	0.126	0.135	+0.009	0.635	0.567	-0.067
65	0.128	0.139	+0.011	0.763	0.706	-0.057
66	0.062	0.050	-0.012	0.824	0.756	-0.069
67	0.020	0.125	+0.105	0.845	0.881	+0.036
68	0.037	0.085	+0.048	0.882	0.965	+0.084
69	0.080	0.023	-0.057	0.962	0.988	+0.026
70	0.038	0.012	-0.026	1.000	1.000	+0.000

Note. The selected calibration uses $\beta = 0.9685$, bequest intensity 1100, bequest shift 200, participation cost 1.05, and participation cost age slope 0.02.

the HRS target. The model also captures the fact that claiming is front-loaded rather than concentrated at age 70.

The remaining weakness is the interior timing of claims. The model places too little mass at age 63, too much mass at ages 67–68, and too little mass at ages 69–70. As a result, cumulative claiming by 65 remains 5.7 percentage points below the HRS target even though the age-62 share and mean claiming age are close. I therefore view the selected calibration as matching the first-order claiming margin well, while leaving some residual mismatch in how delayed claiming is distributed across the post-62 ages.

6.3. Results

7. Counterfactuals

8. Conclusion

A. Further Details on Social Security

A.1. AIME & PIA

Social Security retirement benefits are a concave function of a worker's lifetime covered earnings, summarized in two steps: the Average Indexed Monthly Earnings (AIME) and the Primary Insurance Amount (PIA).

Average Indexed Monthly Earnings (AIME). Let y_k denote nominal Social Security covered earnings in calendar year k, capped at that year's taxable maximum $\bar{y}_k$. Earnings are first wage-indexed to the year in which the worker turns 60. Denoting by AWI_k the national average wage index in year k and by k_{60} the year the worker reaches age 60, indexed earnings are

$$\tilde{y}_k = \min(y_k, \bar{y}_k) \times \begin{cases} \dfrac{\mathrm{AWI}_{k_{60}}}{\mathrm{AWI}_k} & \text{if } k < k_{60}, \\ 1 & \text{if } k \geq k_{60}. \end{cases}$$

The AIME, denoted $\bar{e}$ in the model, is the average of the highest 35 years of indexed earnings, expressed in monthly terms. Letting $\mathcal{H}$ be the set of the 35 highest values of $\tilde{y}_k$,

$$\bar{e} = \frac{1}{12 \times 35} \sum_{k \in \mathcal{H}} \tilde{y}_k.$$

Years with no earnings enter as zeros if the worker has fewer than 35 years of positive earnings, which mechanically lowers the AIME.

The law of motion for the AIME is

$$\bar{e}_{t+1} = \min\left[\bar{e}_t + \max\left(0,\ \tfrac{\min(\omega_t n_t,\ \bar{a}) - \bar{e}_t}{35}\right),\ \bar{a}\right]$$

where $\bar{a}$ is the maximum taxable labor income. For tractability, I assume that an additional year of work replaces a year of earnings equal to the current average $\bar{e}_t$, so a year of earnings above (below) the running average raises (leaves unchanged) the AIME[1]. Then the AIME is converted into the Primary Insurance Amount, $\mathrm{PIA}(\bar{e}_t)$, by a piecewise-linear

[1] Capping the earnings flow at $\bar{a}$ reflects that earnings above the taxable maximum are neither taxed nor credited toward the AIME. The outer cap is kept for clarity. The denominator presumes a full 35-year earnings history, which holds over the older ages modeled here.

function, which the actuarial adjustment $\Gamma(q)$ then scales for the claiming age.

Primary Insurance Amount (PIA). The PIA is a piecewise-linear, concave function of the AIME with two bend points $b_1 < b_2$ and marginal replacement rates that decline across brackets:

$$\text{PIA}(\bar{e}) = 0.90\ \min(\bar{e}, b_1) + 0.32\left[\min(\bar{e}, b_2) - b_1\right]^+ + 0.15\left[\bar{e} - b_2\right]^+,$$

where $[x]^+ \equiv \max(x, 0)$. The replacement rates 90%, 32%, and 15% are fixed by statute, while the bend points b_1 and b_2 are indexed each year to the national average wage index. The declining marginal rates make the benefit formula progressive: a dollar of additional AIME raises the PIA by 90 cents for low earners but only 15 cents for high earners above the second bend point. The PIA is the monthly benefit payable to a worker who claims at the full retirement age; benefits at other claiming ages are obtained by applying the adjustment factors described next.

A.2. Adjustments

The PIA is the benefit payable to a worker who claims exactly at the full retirement age (FRA). Claiming earlier or later rescales the benefit through a statutory adjustment factor, and earnings before the FRA may temporarily withhold part of the benefit through the Retirement Earnings Test. The baseline model uses the common post-transition rule for the 1941–1945 birth cohorts, setting the FRA to age 66 and the delayed retirement credit to 8% per year. Let q denote the claiming age and let $n = q - 66$ be the number of years claiming precedes ($n < 0$) or follows ($n > 0$) the FRA.

Early-claiming reduction. Claiming before the FRA reduces the benefit permanently. Measuring $n_m = 12(q - 66)$ in months, the reduction is $\frac{5}{9}$ of one percent per month for each of the first 36 months before the FRA, and $\frac{5}{12}$ of one percent per month for any additional month beyond 36:

$$\Gamma(q) = 1 - \frac{5}{900}\ \min(|n_m|, 36) - \frac{5}{1200}\ \max(|n_m| - 36,\ 0), \qquad q < 66.$$

With FRA equal to 66, claiming at the earliest eligibility age of 62 gives $\Gamma(62) = 0.75$.

Delayed retirement credits. Delaying past the FRA increases the benefit by the delayed retirement credit (DRC) of 8% per year, accruing until age 70:

$$\Gamma(q) = 1 + 0.08\,n, \qquad 66 \le q \le 70.$$

Claiming at age 70 therefore gives $\Gamma(70) = 1.32$. No further credit accrues after age 70, so $\Gamma(q) = \Gamma(70)$ for $q > 70$. Collecting the two cases, the benefit before any earnings test is $\Gamma(q)\,\mathrm{PIA}(\bar{e})$, where $\Gamma(66) = 1$.

Retirement Earnings Test (RET). Before the FRA, current labor earnings temporarily withhold benefits. Let E_t be annual earnings and $\underline{E}_t$ the annual exempt amount. Benefits are withheld at a rate of \$1 for every \$2 of earnings above the exempt amount in years strictly before the FRA, and at the more lenient rate of \$1 for every \$3 above a higher exempt amount $\underline{E}_t^{\mathrm{FRA}}$ during the year the worker reaches the FRA:

$$\Upsilon_t = \begin{cases} \frac{1}{2}\,\max\bigl(E_t - \underline{E}_t,\,0\bigr) & \text{in years before the FRA,} \\ \frac{1}{3}\,\max\bigl(E_t - \underline{E}_t^{\mathrm{FRA}},\,0\bigr) & \text{in the year of reaching the FRA,} \\ 0 & \text{at and after the FRA.} \end{cases}$$

The withheld amount Υ_t is the term subtracted in the benefit equation in the main text. The RET is not a pure tax: benefits withheld before the FRA are credited back at the FRA through an upward adjustment of the reduction factor, so that, for a worker of average mortality, the test is approximately actuarially neutral. It nonetheless affects the timing of benefit receipt and, because many workers perceive it as a tax rather than as a deferral of actuarially fair benefits, can distort the labor-supply decision of early claimants (Friedberg 2000; Gruber and Orszag 2003; Coile et al. 2002; Gelber, Jones, and Sacks 2020).

B. Appendix: Data

B.1. Summary Statistics

TABLE 4. Summary statistics for HRS samples

	Full HRS sample	Estimation HRS sample
A. Sample size		
Respondents	27,470	10,569
Person-wave observations	194,055	69,182
Observed interview waves	7.1	6.5
Household-head proxy	91.8%	100.0%
B. Demographics at sample entry		
Birth year	1948.7	1949.3
Age	53.5	54.1
Male	44.8%	100.0%
Years of education	12.6	12.8
College	23.4%	26.6%
Married or partnered	74.8%	77.0%
Single/unmarried	25.2%	23.0%
Bad health	26.0%	25.3%
C. Economic variables at sample entry		
Working for pay	70.2%	76.6%
Annual earnings ($000)	28.0	38.5
Household income ($000)	69.5	76.7
Total wealth less IRA ($000)	227.7	252.2
Financial wealth ($000)	52.1	55.8

Note. Full sample includes all RAND HRS respondents born 1933–1965. Estimation sample restricts the same birth cohorts to male respondents who are ever observed as the RAND family or financial respondent, our household-head proxy. Entries are unweighted respondent counts, person-wave counts, means, or shares. Demographic and economic variables are measured at each respondent's first observed HRS interview wave. Dollar values are in thousands of nominal dollars.

Source. RAND HRS 1992–2020 Version 2.

B.2. Measuring Insolvency Index

The Social Security insolvency index is constructed from a real-time monthly panel of official solvency information. The raw data come from SSA Trustees Reports, CBO Social Security baseline documents and testimony, congressional testimony, and public monthly macro series. The unit of observation is a month. I build the index on the OASI basis, the legal headline series used for the trust-fund accounting and the relevant series for the household benefit in the model. Official annual or irregularly released objects are dated by their release month and carried forward until the next release, so the index uses only information available as of each month.

Table 5 lists the data variables and the moments used to construct the index. Each moment is signed so that a larger value means a more adverse solvency environment. The stress block captures the level of the financing problem, while the uncertainty block captures dispersion, disagreement, revisions, and official discussion of solvency uncertainty.

I convert these moments into the index in four steps, all on the OASI basis. First, I orient every input so that higher values correspond to greater insolvency risk. For example, the actuarial-balance component is multiplied by negative sign, years to depletion enters with a negative sign, and the payable-benefit component is measured as the shortfall from full scheduled benefits. Let m_{jt} denote oriented moment j in month t. Each moment is standardized,

$$x_{jt} = \frac{m_{jt} - \overline{m}_j}{s_j},$$

where $\overline{m}_j$ and s_j are computed from the non-missing release-sample observations for that moment. This puts actuarial balances, depletion horizons, trust-fund ratios, text rates, and macro series on a common scale.

Second, I group the standardized moments into a stress block and an uncertainty block. The stress block is

$$\mathcal{S}_t = \{-AB_t,\ UO_t,\ -YTD_t,\ 100 - PAY_t,\ TFRW_t,\ MPBS_t\},$$

where *AB* is the 75-year actuarial balance, *UO* the open-group unfunded obligation, *YTD* years to depletion, *PAY* the payable-benefit ratio at depletion, *TFRW* trust-fund-ratio weakness, and *MPBS* macro payroll-base stress. The uncertainty block is

$$\mathcal{U}_t = \{SW_t,\ STW_t,\ RV_t,\ D_t,\ MRR_t,\ TXT_t\},$$

where *SW* is the Trustees scenario width, *STW* the stochastic width, *RV* cross-vintage revision volatility, *D* the absolute SSA–CBO disagreement in years to depletion, *MRR* macro revision risk, and *TXT* the supervised official-text uncertainty signal. The text signal enters only the uncertainty block and only after satisfying the pre-specified text

TABLE 5. Variables and moments used in the insolvency-risk index

Source block	Data variables	Moment entering the index
SSA Trustees Reports	75-year actuarial balance as a percent of taxable payroll	Stress: negative actuarial balance, so a larger funding deficit raises the index
SSA Trustees Reports	Open-group unfunded obligation, reported in trillions of dollars and as shares of taxable payroll and GDP	Stress: larger unfunded obligation
SSA Trustees Reports and CBO	Projected depletion year and years to depletion for OASI	Stress: shorter years to depletion; uncertainty: absolute SSA–CBO gap in years to depletion
SSA Trustees Reports	Payable-benefits ratio at trust-fund depletion	Stress: payable-benefit shortfall, defined as 100 minus the payable ratio
SSA Trustees Reports	Trust-fund-ratio summaries, including 10-year and 25-year minimum ratios	Stress: trust-fund-ratio weakness, defined from the shortfall of the reserve ratio below 100 percent
SSA Trustees Reports	Intermediate, low-cost, high-cost, and stochastic projections of depletion timing	Uncertainty: scenario-width and stochastic-width measures of depletion-year dispersion
SSA Trustees Reports	Release-to-release revisions in actuarial balance, unfunded obligation, years to depletion, and scenario width	Uncertainty: rolling volatility of cross-vintage revision shocks
Public macro series	Payroll employment (PAYEMS), average hourly earnings (AHETPI), and the unemployment rate (UNRATE)	Stress: payroll-base stress from weak wage-bill growth and high unemployment; uncertainty: macro revision risk from changes and volatility in payroll-base stress
Official text corpus	SSA, CBO, and congressional sentences mentioning Social Security, trust funds, solvency, depletion, payable benefits, reforms, or uncertainty	Uncertainty: supervised monthly uncertainty rate, after the text gate approves the signal for the uncertainty block

Note. These inputs feed the OASI solvency-risk index plotted in Appendix Figure 3.

Source. Author's calculations from SSA Trustees Reports, CBO reports and testimony, congressional testimony, and FRED public macro series.

gate.

Third, I estimate one latent factor for each block. For block $k \in \{S, U\}$, the measurement system is

$$x_{jt}^k = \lambda_j^k F_t^k + \varepsilon_{jt}^k, \qquad F_t^k = \phi^k F_{t-1}^k + \eta_t^k.$$

The factor is estimated with a Kalman smoother, which permits missing components when a source is unavailable in a given month. The stress factor is sign-normalized to load positively on the adverse actuarial-balance component; the uncertainty factor is sign-normalized to load positively on scenario width. Thus higher F_t^S and higher F_t^U both mean greater insolvency risk.

Finally, I standardize the two estimated factors over the pre-COVID baseline period 1995–2019 and combine them into the headline solvency state:

$$\widetilde{S}_t = \frac{F_t^S - \mu_{95-19}^S}{\sigma_{95-19}^S}, \qquad \widetilde{U}_t = \frac{F_t^U - \mu_{95-19}^U}{\sigma_{95-19}^U},$$

$$G_t = 0.70\,\widetilde{S}_t + 0.30\,\widetilde{U}_t.$$

The fixed weights place most weight on the measured fiscal stress of the program, while allowing uncertainty, disagreement, revisions, and official text to shift the perceived risk state. The plotted index scale in Appendix Figure 3 is a final normalization of this headline state,

$$I_t = 100 + 10 \times \frac{G_t - \mu_{95-19}}{\sigma_{95-19}},$$

so that I_t has mean 100 and standard deviation 10 over 1995–2019. The main text then maps the annual OASI solvency state G_t into the subjective annual benefit-risk probability shown in Figure 1.

The mapping from the index to the respondent's subjective probability is estimated using beliefs about Social Security from the HRS Internet Surveys. I use the general Social Security expectations item in the 2006, 2007, and 2011 surveys. The question asks respondents, thinking of the Social Security program in general and not just their own benefits, for the percent chance that Congress will change Social Security sometime in the next 10 years so that it becomes less generous than now. The wording is held essentially fixed across the three waves. In 2011 this item was fielded in the Collection A arm of a randomized wording experiment.

I interpret the response as a proxy for subjective belief about a future policy-induced reduction in Social Security generosity, not as an objective forecast of trust-fund depletion. Responses are reported on a 0–100 scale. I retain valid responses between 0 and 100, divide them by 100 to construct $p_{it}^{HRS} \in [0, 1]$, and set all other values to missing. Because the 2011 item was part of a randomized wording experiment, the response is observed only for respondents assigned to Collection A.

The Internet Surveys also contain more personalized items that can be used for robustness. In 2006 and 2007, current Social Security recipients were asked for the chance that the benefits they were currently receiving would be cut sometime over the next 10 years. Respondents who were not currently receiving Social Security were instead asked for the chance that changes over the next 10 years would reduce their future benefits relative to what they would receive under the current system. In 2011, a personalized follow-up asks whether the previously described Social Security changes might affect the respondent's own benefits. This item is conditional on a positive response to the preceding general-cut question and, for non-recipients, a positive probability of ever receiving Social Security; it is also restricted to respondents younger than 70. The 2011 survey additionally randomized respondents to an alternative general-belief wording, which asks for the chance that Social Security benefits will be cut in the next 10 years. This alternative wording is fielded in place of, rather than in addition to, the general item described above. I use the personalized items and the alternative-wording item only for robustness checks. The preferred calibration uses the general 2011 item because its wording is most directly comparable with the general 10-year item used in 2006 and 2007.

The model uses an annual perceived benefit-risk probability,

$$p_t = \Lambda(\alpha_0 + \alpha_1 G_t), \qquad \Lambda(z) = \frac{\exp(z)}{1 + \exp(z)}.$$

The HRS question, however, asks respondents for a 10-year probability that Social Security will become less generous. Therefore the annual probability implied by the index is compared with the survey response only after converting it into the corresponding 10-year probability,

$$P_t^{10}(\alpha_0, \alpha_1) = 1 - [1 - \Lambda(\alpha_0 + \alpha_1 G_t)]^{10}.$$

This distinction is important: the survey response is used on its original 10-year probability scale, while the object entering the annual household problem is the period-specific probability p_t consistent with that 10-year belief. The model does not compound this object into an absorbing reform state; households instead update p_t with the calendar-year belief path.

Let p_{it}^{HRS} denote respondent i's reported 10-year probability in survey year t, and let w_{it} be the nearest-prior RAND HRS respondent weight. The preferred specification estimates (α_0, α_1) by weighted nonlinear least squares,

$$(\widehat{\alpha}_0, \widehat{\alpha}_1) = \arg\min_{\alpha_0, \alpha_1} \frac{\sum_i w_{it} \left[p_{it}^{HRS} - P_t^{10}(\alpha_0, \alpha_1) \right]^2}{\sum_i w_{it}},$$

pooling the HRS Internet Survey waves with the standard general Social Security wording in 2006, 2007, and 2011. The estimation sample requires a non-missing 10-year belief and a positive prior respondent weight. It does not condition on age, sex, claiming status, or cohort in this step.

Identification comes from the time-series relationship between aggregate solvency conditions and aggregate subjective beliefs. Since G_t is common to all respondents in a survey year, the two parameters are identified by the weighted wave-level means of HRS beliefs across the three survey years. The intercept α_0 pins down the average perceived risk level, while α_1 is identified by how perceived 10-year risk changes as G_t moves across survey years. Individual observations determine the weighted moments and the respondent-cluster bootstrap precision, but not additional cross-sectional slopes.

TABLE 6. HRS moments used to estimate the subjective-probability mapping

Survey year	G_t	Observations	HRS 10-year risk	Model 10-year risk	Model annual p_t
2006	-0.635	1,208	63.3%	60.9%	9.0%
2007	-0.467	1,487	60.3%	62.2%	9.3%
2011	0.865	2,180	72.1%	72.0%	11.9%

Note. HRS moments are weighted means of the standard 10-year "less generous" Social Security belief question. The fitted values use $P_t^{10} = 1 - (1 - p_t)^{10}$, where $p_t = \Lambda(\alpha_0 + \alpha_1 G_t)$.

Source. Author's calculations from the HRS Internet Survey and the annual solvency-risk index.

The preferred estimates are

$$\widehat{\alpha}_0 = -2.182 \quad (0.014), \qquad \widehat{\alpha}_1 = 0.212 \quad (0.017),$$

where parentheses report respondent-cluster bootstrap standard errors based on 250 replications. The positive slope implies that a higher solvency-risk state raises the perceived probability of a benefit adjustment. At $G_t = 0$, the annual perceived probability is 10.1 percent, which corresponds to a 10-year risk of 65.7 percent. Applying the same mapping to the annual index gives the subjective annual-probability series plotted in Figure 1; for example, the 2026 value $G_t = 3.076$ implies an annual subjective probability of 17.8 percent.

B.3. Solvency index scale

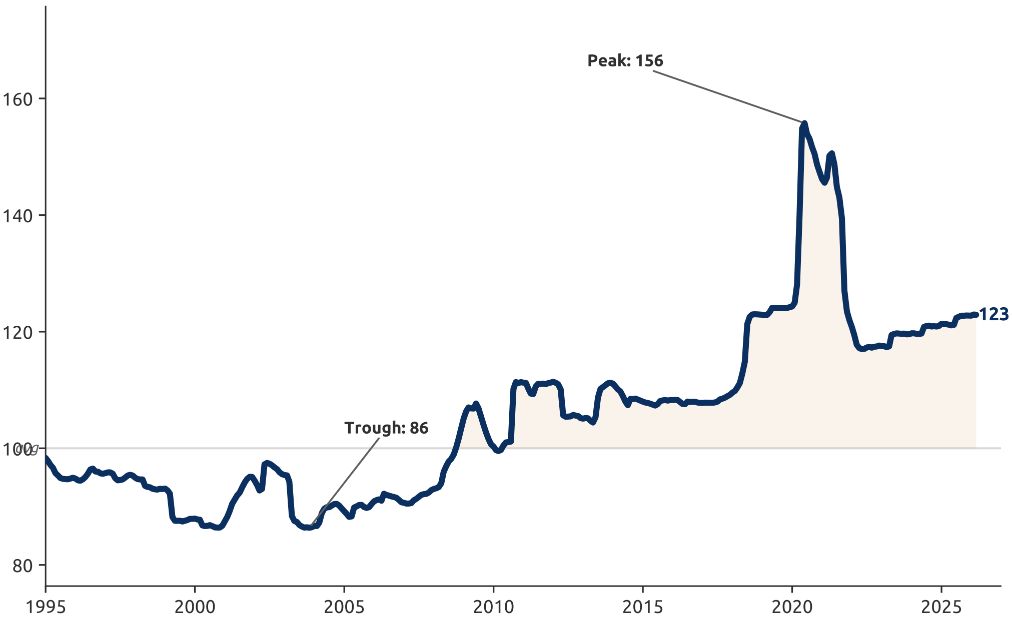


FIGURE 3. Social Security solvency-risk index, OASDI

Note. This figure reports the underlying OASDI solvency-risk index before conversion to the subjective annual probability in Figure 1. The index combines official solvency fundamentals and validated official text signals, with the 1995–2019 average normalized to 100. Higher values indicate a more adverse Social Security solvency environment. Source: author's calculations from SSA Trustees Reports, CBO reports and testimony, congressional testimony, and public macro series.

C. Appendix: Estimation

C.1. First-Stage Parameters

The first stage fixes the household's economic and institutional environment and holds it constant while the preference parameters are estimated in the second stage (Table 9). Table 8 collects the first-stage parameters: the wage process is estimated from the PSID, health and medical-expense objects are estimated from MEPS and HRS panel data, the Social Security parameters use a common rule approximation for the 1941–1945 birth cohorts, and the insolvency-belief mapping is calibrated to HRS Internet Survey expectations (Appendix B).

C.2. Wage Process

The wage process has a deterministic component that varies by fixed type and a persistent stochastic component. Both objects are estimated from the PSID for male heads/reference persons ages 25–80 and are used to discipline the working-age and older-age wage process in the model. HRS wage estimates are produced separately for ages 50–80 as an older-age diagnostic, but the current production wage parameters in Table 8 use the PSID estimates.

The estimation sample keeps male heads/reference persons between ages 25 and 80 in the annual PSID family files. I use the downloaded family-file archives from survey years 1977–2019, which provide wage years 1976–2018; earlier family files are not used in this version because they do not consistently contain the annual head-hours measure needed to construct hourly wages. For survey years 1985–1992, the family files report the 1968 family identifier and person number of the householder. I use this exact person key when the householder is also the family-unit head. Outside those records, I construct an approximate head key from the family identifier, sex, and implied birth cohort. This approximation is sufficient for the deterministic profile but is treated cautiously when forming the balanced wage-shock panel.

The hourly wage is constructed as annual labor earnings divided by annual hours worked. Annual labor earnings and annual hours are selected from each family file by label, prioritizing total labor income and total annual hours for the head/reference person in the prior calendar year. Labor-force participation for the selection equation is based on annual hours above the employment threshold or, when hours are missing, positive annual labor earnings. Annual hours below 250 are treated as nonemployment. The wage is converted to 2020 dollars with annual CPI-U. Valid wage observations require positive earnings, annual hours between 250 and 5,000, and a real hourly wage between \$3.50 and \$75.00. The baseline PSID family-file wage sample contains 118,944 valid wage observations; after dropping people without a complete fixed type and applying the wage-equation covariate filters, 86,144 typed wage observations for 8,025 persons enter the wage regression. All pooled weighted moments and regressions use PSID weights normalized to mean one within wage year, preserving cross-sectional relative weights while preventing later waves with larger population weights from dominating the pooled estimates.

Education is fixed at the individual level using the maximum observed completed schooling in the extract. The estimation uses two education groups: less than college, for fewer than 16 completed years, and BA+, for at least 16 completed years. Permanent productivity type is then defined from the full sample, independent of education, using terciles of each person's first observed valid real log hourly wage. The fixed type is the cross of the two education groups and the three productivity terciles, giving six wage types: less-than-college/BA+ crossed with low, middle, and high productivity. People without both education and productivity type are dropped from the wage-process sample. Health is not crossed into the PSID wage fixed types because the family-file wage panel does not consistently observe head/reference-person health status.

Let D_{it} denote labor-market participation and let $a_{it} = age_{it} - 50$. To diagnose selection into observed wages, I estimate a weighted participation probit,

$$\Pr(D_{it} = 1) = \Phi\Big[\alpha_0 + \alpha_1 a_{it} + \alpha_2 a_{it}^2 + \alpha_3 Coll_i + \alpha_4 Coll_i a_{it} + \alpha_5 Coll_i a_{it}^2 \\ + \alpha_6 Marr_{it} + \alpha_7 Marr_{it} a_{it} + \alpha_8 Child_{it} + \alpha_9 Child_{it} a_{it} + \delta_t\Big].$$

where $Coll_i$ is the college indicator, $Marr_{it}$ is an indicator for being married, $Child_{it}$ is an indicator for children in the family unit, and δ_t are calendar-year fixed effects. The inverse Mills ratio λ_{it} is computed from this first-stage probit for participating observations.

The baseline deterministic component is estimated on working observations with valid wages using weighted least squares,

$$\log \omega_{it} = \mu_{\vartheta(i)} + \beta_1 a_{it} + \beta_2 a_{it}^2 + \beta_3 Marr_{it} + \beta_4 Marr_{it} a_{it} + \delta_t + u_{it},$$

where $\mu_{\vartheta(i)}$ is a fixed-type intercept for the six education-by-productivity wage types and δ_t are year fixed effects. The fixed-type intercepts absorb the level differences by education and permanent productivity. The age profile is restricted to have common slope and curvature across fixed types, so productivity and education shift the deterministic profile through the fixed-type intercepts.[2] The regression therefore uses year fixed effects, not birth-cohort fixed effects. This choice removes aggregate calendar-year wage conditions from the profile; unrestricted birth-year fixed effects are not included because age, year, and birth year are mechanically related. The inverse Mills ratio is not included in the baseline wage-profile equation. The deterministic wage component used in the model and in the residual construction is

$$\widehat{w}_{it} = \widehat{\mu}_{\vartheta(i)} + \widehat{\beta}_1 a_{it} + \widehat{\beta}_2 a_{it}^2 + \widehat{\beta}_3 Marr_{it} + \widehat{\beta}_4 Marr_{it} a_{it} + \widehat{\delta}_t.$$

The wage residual used for the stochastic process is

$$\widehat{u}_{it} = \log \omega_{it} - \widehat{w}_{it}.$$

The solver input reports age-by-type wage profiles with the year effects evaluated at

[2] A separate college main effect is not included because fixed types are already defined as education-by-productivity pairs, so a college level effect is absorbed by the fixed-type intercepts.

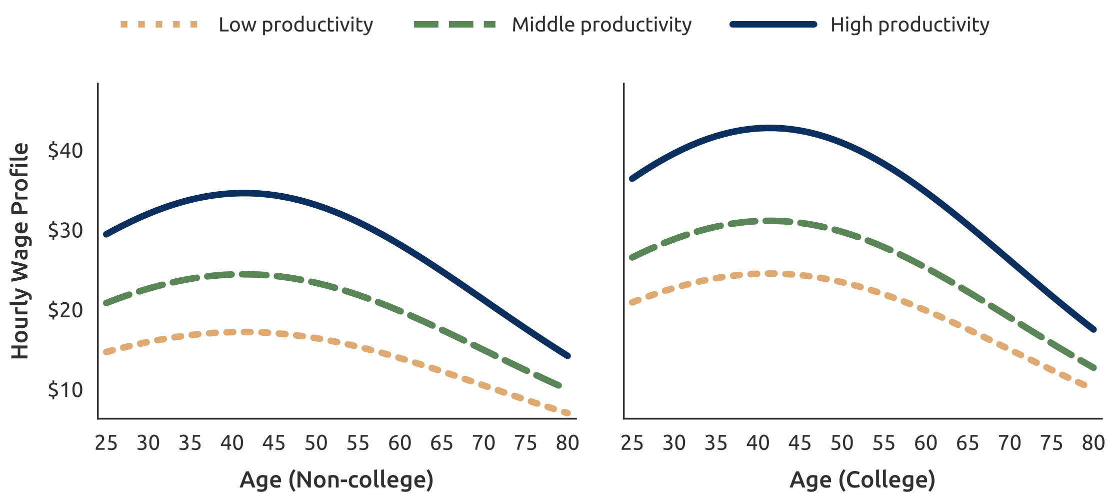


FIGURE 4. PSID hourly wage profiles by education and productivity type

Note. The figure plots deterministic hourly wage profiles for male PSID heads/reference persons ages 25–80. The left panel plots less-than-college fixed types and the right panel plots BA+ fixed types. Hourly wages are annual labor earnings divided by annual hours worked and converted to 2020 dollars using CPI-U annual averages.

their sample-weighted mean and marital status evaluated at the type-specific weighted mean.

Figure 4 shows the resulting deterministic profiles side by side for less-than-college and BA+ fixed types, using a common vertical scale. The model wage offer is

$$\omega_t = \chi(\iota_t) \exp\{w_t(\theta, e) + \xi_t\},$$

where unemployment ι_t is an i.i.d. shock observed at the start of the period and is not carried forward as a state. In the current wage-process estimate, $\chi(0) = \chi(1) = 1$, so unemployment does not directly depress the hourly wage. The stochastic component is estimated from the centered PSID residuals $\widehat{u}_{it}$. The parameters ρ and σ_ζ^2 are chosen by simulated moments. Following the spirit of French (2005), the wage-shock moments are computed on a balanced annual panel. With the PSID Individual File link, the direct annual wage measure supports the French target wage years 1977–1986. This panel contains 1,410 persons and 14,100 person-year observations. The simulator keeps this balanced person-year structure and simulates 100 antithetic residual panels using common random numbers. The weighting matrix is the identity. The data moments are the weighted residual variance and weighted residual autocovariances at lags one, two, and three:

$$m = \left(\mathrm{Var}(\widehat{u}_{it}), \mathrm{Cov}(\widehat{u}_{it}, \widehat{u}_{i,t-1}), \mathrm{Cov}(\widehat{u}_{it}, \widehat{u}_{i,t-2}), \mathrm{Cov}(\widehat{u}_{it}, \widehat{u}_{i,t-3})\right)'.$$

In the PSID sample these moments are

$$(0.1575,\ 0.1091,\ 0.0961,\ 0.0863)'.$$

The corresponding pair counts are 12,690 one-year pairs, 11,280 two-year pairs, and 9,870 three-year pairs. A pure AR(1) specification gives $\widehat{\rho} = 0.804$ and $\widehat{\sigma}^2_\zeta = 0.0530$. Because hourly wages contain substantial transitory and measurement variation, the preferred MSM specification allows an auxiliary transitory residual component,

$$\widehat{u}_{it} = \xi_{it} + \eta^{\omega}_{it}, \qquad \eta^{\omega}_{it} \sim N(0, \sigma^2_{\eta^\omega}),$$

which is not carried forward as a model state. This specification matches the lagged autocovariances more closely and yields $\widehat{\rho} = 0.889$, $\widehat{\sigma}^2_\zeta = 0.0257$, and $\widehat{\sigma}^2_{\eta^\omega} = 0.0350$.

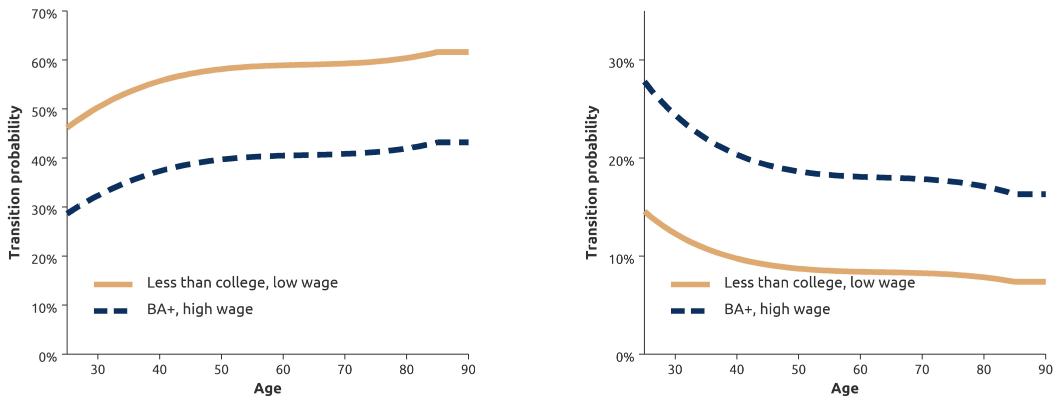
C.3. Health and Survival Transition

Health transition. The health transition matrix is estimated from the Household Component of the Medical Expenditure Panel Survey (MEPS-HC), using the pooled full-year consolidated files from 1996 through 2023. The estimation sample includes individuals ages 25–85 with nonmissing self-reported health, education, annual wage income, and final person weights. The raw five-category self-reported health measure is collapsed into three states: good health equals excellent health, fair health equals very good or good health, and bad health equals fair or poor health. A transition is measured within each annual MEPS file as the move from the first observed self-reported health measure in the year to the last observed self-reported health measure in the year.

Permanent type is defined by education and a productivity proxy. Education has two groups: less than college and BA+, where BA+ is measured as at least 16 completed years. Since annual wage income is available in every MEPS year used here, I proxy productivity by the pooled weighted tercile of CPI-adjusted annual wage income, measured in 2023 dollars. The resulting tercile cutoffs are \$11,322 and \$52,650. Thus the health process is estimated for six education-by-productivity types. The final person weights are used in estimation and normalized to have mean one.

To smooth sparse age-by-type cells, I estimate the transition probabilities with a pooled weighted ordered probit. Let h'_i denote next health, ordered from good to fair to bad, and let higher values of the latent index correspond to worse future health. The latent index includes indicators for current fair and bad health, a cubic polynomial in age, and indicators for the six education-by-productivity types, with less-than-college and low-wage individuals as the omitted type. The fitted ordered probit is then used to predict a 3×3 transition matrix for each age, education group, and productivity group. Public-use MEPS top-codes age at 85, so the estimated age-85 transition matrix is used for all model ages 86 and older.

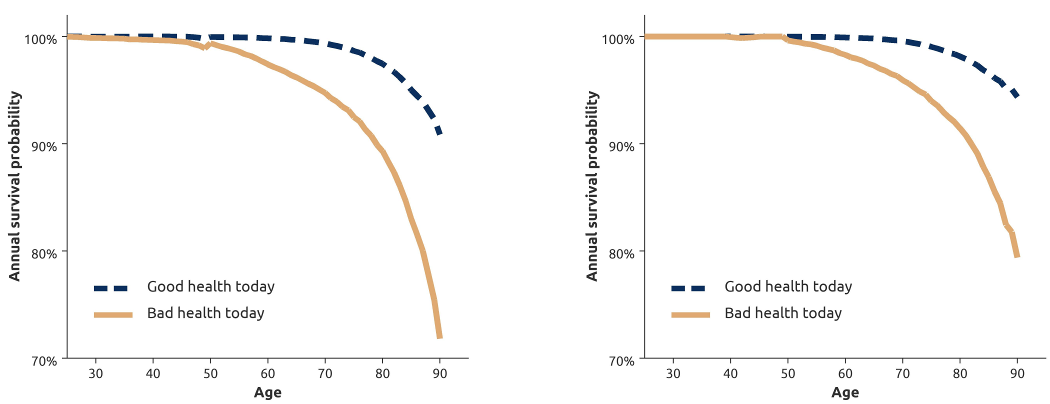
Survival Probabilities. Objective survival probabilities are estimated with the same collapsed lagged health states and the same education-by-productivity fixed types, but use the data source appropriate for the model age. For ages below 50, I use MEPS-HC annual records and define death from the final-round person disposition codes for deceased respondents. For ages 50 and older, I use RAND HRS wave-to-wave records and define death from the next-wave interview status. Since HRS interviews are biennial, the fitted HRS death probability is converted to an annual survival probability under a constant within-interval hazard. The survival equation is a weighted binary ordered probit, with the death indicator ordered above survival. The latent index includes lagged health status interacted with a cubic polynomial in age; fixed type, marital status, and gender interacted with the same age polynomial; and birth-year cohort fixed effects. Predicted survival probabilities average the cohort fixed effects using the source- and fixed-type-specific weighted birth-year distribution at each age. To avoid extrapolating sparse very-old-age HRS cells, I estimate and plot survival probabilities through age 90; for model ages 91–99, the age-90 survival probabilities are used for the same health state, fixed type, marital status, and gender.



A. Good → Fair or Bad Health B. Fair → Good Health

FIGURE 5. Health transition probabilities by age and type

Note. The figure reports fitted health transition probabilities from a pooled weighted ordered probit estimated on MEPS-HC full-year consolidated files from 1996 through 2023. The estimation sample includes individuals ages 25–85 with nonmissing self-reported health, education, annual wage income, and final person weights. Panel A plots the probability of moving from good health today to fair or bad health next period; Panel B plots the probability of recovering to good health next period conditional on fair health today.



A. Less-than-college, low-wage type B. BA+, high-wage type

FIGURE 6. Annual survival probabilities by age and health

Note. Each panel fixes an education-by-productivity type and compares predicted annual survival probabilities for individuals in good versus bad health at the start of the year. The left panel is the less-than-college, low-wage type and the right panel is the BA+, high-wage type. Education is grouped into less than college and higher than college.

C.4. Health Insurance Type

Health insurance type is assigned from RAND HRS employer retiree-coverage information for the main 1941–1945 birth cohorts. The sample is restricted to male respondents who are ever observed as the RAND family or financial respondent, matching the household-head proxy used elsewhere in the HRS analysis. In the current computational specification, insurance type is not a separate fixed type solved in the dynamic program, and the production medical-expense process is not split by employer health-insurance category. Instead, the assignment below is used to initialize the empirical labor-history state and to document retiree, tied, and no-coverage shares within each of the six education-by-productivity fixed types. To document the retiree-coverage classification, I use only observations before Medicare eligibility and inspect the first informative observation at ages 62–64.

The variable summarizes whether the respondent's own employer-provided health insurance would cover the respondent in retirement. Respondents are assigned to the retiree type only when the plan covers the respondent both to and after age 65. Plans that do not cover retirement, cover only to age 65, or cover to age 65 with unknown post-65 coverage are classified as tied. These plans may bridge the worker to Medicare, but they do not provide the lifetime retiree coverage represented by the absorbing retiree state. Respondents with no respondent employer-provided insurance are assigned to the none type. Other special missing values are not used for the type assignment.

TABLE 7. Health insurance type from RAND HRS retiree-coverage reports

Insurance type	RAND HRS assignment rule	Persons	Share (%)
Retiree	HERET=3 or 4	214	35.3
Tied	HERET=0, 1, or 2	190	31.3
None	HERET=.n	203	33.4
Total informative	First informative age 62–64 observation	607	100.0

Note. The table uses RAND HRS 1992–2020 Version 2 and the main 1941–1945 birth cohorts. The sample keeps male respondents who are ever observed as the RAND family or financial respondent. Of 1,133 respondents in this cohort-sample, 810 are observed at ages 62–64 and 607 have an informative value at those ages. The reported distribution uses each respondent's first informative observation between ages 62 and 64. HERET=3 indicates coverage to and after age 65; HERET=4 is the early-wave retiree-coverage summary code. HERET=0, 1, and 2 are classified as tied because they do not imply absorbing retiree coverage after age 65. Special missing values other than .n are excluded.

C.5. Medical Expenditure

The medical-expense profiles are estimated from RAND HRS 1992–2020 Version 2 for the male household-head proxy sample used in the HRS analysis. The sample keeps respondents born between 1933 and 1965, observed at ages 55–84, with nonmissing medical expenses, health status, labor-force participation, and health-insurance classification. The estimation sample contains 54,399 person-waves for 9,589 respondents. The fixed-type medical profiles use the 7,047 respondents who can be linked to the HRS wage-panel education-by-productivity type assignment.

The HRS reports household out-of-pocket medical spending rather than total medical spending paid by all payers. I therefore construct two measures. The first is annual respondent-plus-spouse out-of-pocket spending, converted to 2020 dollars using CPI-U and floored at $250 before taking logs. The second, used for the profiles reported here, adds the French-Jones Medicaid payment imputation to Medicaid households. Specifically, I add $15,877 in 2020 dollars (equivalent $10,000 in 1998 dollars), to the out-of-pocket measure for households in which the respondent or spouse reports Medicaid coverage. Thus the total measure is not observed total spending; it is HRS out-of-pocket spending augmented by the Medicaid payment imputation used to make the HRS measure more comparable to the medical-spending object in French and Jones (2011).

For each education-by-productivity fixed type τ, I estimate the deterministic profile from

$$\log M_{it} = \alpha_\tau + \sum_{k=1}^{3} \beta_{\tau k}(a_{it} - 65)^k + \sum_{s \in \{\text{fair,bad}\}} \sum_{k=0}^{3} \gamma_{\tau s k} \mathbf{1}\{h_{it} = s\}(a_{it} - 65)^k$$
$$+ \lambda_\tau n_{it} + \psi_{\tau, y(t)} + u_{it},$$

where M_{it} is total medical spending, a_{it} is age, h_{it} is the health state, n_{it} is labor-force participation, and $\psi_{\tau, y(t)}$ are year effects. Health follows the same three-state collapse used above: good health is excellent self-reported health, fair health is very good or good self-reported health, and bad health is fair or poor self-reported health. The plotted deterministic mean profile evaluates health and labor-force participation at their type-specific sample means and averages the year effects using the type-specific sample-year distribution. For the first full model run, the solver uses lognormal medical-spending cells by age band, health, labor-force participation, and retiree/tied/no-coverage environment. Because employer health-insurance type is not a solved fixed type, the Fortran solver forms the expected medical cost within each education-by-productivity type using the

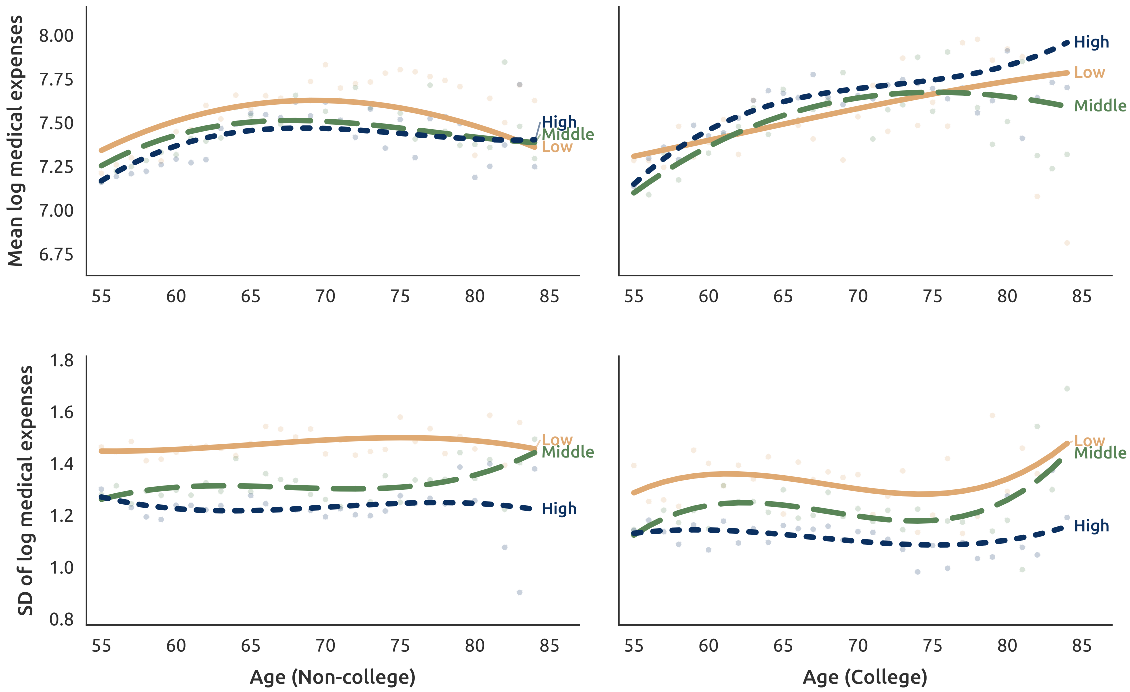


FIGURE 7. Mean and dispersion of log total medical expenses by age and fixed type

Note. The upper row plots deterministic mean profiles of log total medical expenses, and the lower row plots exact-age standard deviations of log total medical expenses with weighted cubic smooths in age. Lines are estimated separately by education-by-productivity fixed type. Total medical expenses equal annual respondent-plus-spouse out-of-pocket expenses plus a Medicaid payment imputation for Medicaid households. Out-of-pocket expenses are annualized, converted to 2020 dollars using CPI-U, and floored at $250 before taking logs. The Medicaid imputation is $10,000 in 1998 dollars, equal to $15,877 in 2020 dollars. Source: RAND HRS 1992–2020 Version 2.

empirical age-55 insurance shares described above.

Figure 7 reports both the deterministic mean profile and the age-by-type dispersion of the same log total medical-expense measure. For each fixed type and exact age, I compute the cross-sectional standard deviation of log total medical expenses and omit exact-age cells with fewer than 10 observations. The SD smooth is a cubic polynomial in age, centered at age 65, estimated separately by fixed type and weighted by the square root of the number of observations in the exact-age cell.

The mean profiles rise with age for several fixed types, but the increase is not uniform across the type distribution. The BA+, high-productivity type has the steepest mean profile over much of the age range, while the less-than-college, low-productivity mean profile flattens and declines at older ages. The dispersion of medical expenses is large for every fixed type. The smoothed standard deviation of log total medical expenses is generally between 1.1 and 1.5, so the residual risk around the deterministic profile is economically important. Dispersion is highest for the low- and middle-productivity less-than-college types and lowest for the BA+, high-productivity type. Old-age movements should be read with some caution because exact-age cells become smaller. Overall, the figure supports treating medical-expense risk as age-varying and type-specific rather than as a common variance applied to all households.

TABLE 8. First-stage parameters

Parameter	Description	Value	Source
A. Wage process			
$w_t(\theta, e)$	Deterministic log-wage profile	—	PSID
ρ	AR(1) persistence of the log-wage shock	0.889	PSID
σ_ζ	SD of the persistent wage-shock innovation	0.160	PSID
σ_η^ω	SD of the transitory wage residual (MSM)	0.187	PSID
$\chi(1)$	Wage penalty in an unemployment year	1.00	Bairoliya and McKiernan (2023)
π^ι	Annual unemployment probability	0.056	Bairoliya and McKiernan (2023)
B. Health and mortality			
Π_t^h	Health Markov transitions	—	MEPS
$\pi_t(\theta, e)$	Annual survival, by age, health, and type	—	MEPS, HRS
C. Medical expenses			
$m_t(\theta, e)$	Mean profile of log total medical expense	—	HRS
$\sigma_t(\theta, e)$	SD profile of log total medical expense	—	HRS
ρ_m	AR(1) persistence of the medical shock	0.925	French and Jones (2004)
σ_η^2	Variance of the persistent medical innovation	0.04811	French and Jones (2004)
σ_ε^2	Variance of the transitory medical shock	0.6668	French and Jones (2004)
D. Health insurance			
s^{IC}	Employer-coverage shares $\{R, T, N\}$	Table 7	HRS
t^{med}	Medicare eligibility age	65	Statute
E. Social Security (1941–1945 birth cohorts)			
q	Claiming ages observed in sample	62–70	
τ_{ss}	Payroll tax rate	6.2%	Statute
λ^T	Non-OASI income-tax level at $\bar{z}^T$	0.90	Heathcote, Storesletten, and Violante (2017)
τ^T	Non-OASI income-tax progressivity	0.18	Heathcote, Storesletten, and Violante (2017)
$\bar{z}^T$	Income-tax reference income	50.0	Set
χ^{ss}	Taxable share of Social Security benefits	0.50	Set
r^F	OASI trust-fund portfolio return	7.7%	SSA
$\bar{y}_t^{ss}$	Taxable maximum earnings		Statute
FRA	Full retirement age	66	Statute
$\Gamma(q)$	Delayed credit; early reduction	8%	Statute
PIA	Replacement rate at PIA bend points	90/32/15%	Statute
Υ_t	RET withholding	1/2, 1/3	Statute
F. Insolvency beliefs			
α_0, α_1	Logistic map from solvency index to p_t	$-2.182/0.212$	HRS
δ	Benefit-cut severity, adverse draw	0.21	SSA

Note. For the 1941–1945 cohorts, the baseline uses FRA = 66 and DRC = 8%, treating the small 1941–1942 transition differences as negligible. *Source.* PSID, MEPS, HRS, SSA program rules, HRS Internet Survey expectations; see Appendix B for the belief mapping.

C.6. Second-Stage Parameters

The preference parameters are estimated in the second stage by the method of simulated moments, holding the first-stage parameters of Table 8 fixed. Table 9 collects the structural parameters; the risk-free return, time endowment, and consumption floor are set externally.

TABLE 9. Second-stage parameters

Parameter	Description	Value	Source
A. Preferences			
γ	Relative risk aversion	2.50	SMM
α	Consumption share	0.60	SMM
β	Annual discount factor	0.9685	SMM
B. Bequest			
θ_b	Bequest intensity	1,100	SMM
$\underline{a}$	Bequest shifter (luxury good)	200.0	SMM
η_ϑ	Fixed-type bequest curvature multiplier	(0.75, 1.00, 0.75); (0.85, 0.90, 0.85)	SMM
C. Fixed costs of work			
$\phi_{n,t}$	Participation cost (Int, Slope)	(1.05; 0.02)	SMM
$\phi_{h,t}$	Health-related time cost	0.02	SMM
ϕ_{RE}	Re-entry cost after nonemployment	0.05	SMM
D. Technology (set)			
r	Risk-free real interest rate	0.03	French and Jones (2011)
$\bar{\ell}$	Annual time endowment	2.25 (4,500 hours)	Bairoliya and McKiernan (2023)
$\underline{c}$	Consumption floor	1.0	Set
E. Initialization (set)			
Φ_{55}	Age-55 initial distribution	1941–1945	HRS
$\{\mu_c\}_{1941}^{1965}$	OLG birth-cohort population masses	Age-55 entry years	Census
F_{1996}/G_{1996}	Initial trust-fund ratio	1.93	1996 Trustees Report
G_{1996}	Legacy OASI outlay anchor	1996 OASI cost	SSA OACT
Ω_{1996}	Legacy claimant population weights	OASI stock	SSA
t^0 target	OASI depletion-year validation target	2033	2025 Trustees Report

Note. The table reports the selected second-stage values from the full 1941–1945 calibration run. The discount factor, participation fixed cost, bequest intensity, bequest shifter, and bequest curvature multipliers are calibrated to employment, late-age saving-distribution, and claiming-age validation moments. Fixed-type curvature multipliers are ordered as less-than-college low, less-than-college middle, less-than-college high, BA+ low, BA+ middle, and BA+ high. Monetary quantities are in thousands of 2020 dollars.

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